

§ 176. The real field H_m contained in Ω_m .

Every number of the field Ω_m is transformed by the substitution (r, r^{-1}) into the conjugate-imaginary number which is also obtained by the interchange $(i, -i)$. The product of two such conjugate-imaginary numbers is the square of the absolute value of either of them.

A number in Ω_m is real if it remains unchanged under the interchange (r, r^{-1}) , and only under this condition. Every such number can be expressed rationally by the two-term period $r + r^{-1}$, and therefore belongs to a real field of degree $\frac{1}{2}\varphi(m)$ which is a divisor of Ω_m . This field we shall call the real field contained in Ω_m and denote it by H_m (although other real fields are also contained in Ω_m , namely all divisors of H_m).

The $\frac{1}{2}\varphi(m) = \frac{1}{2}\mu$ numbers

$$1, \quad r + r^{-1}, \quad r^2 + r^{-2}, \dots, \quad r^{\frac{1}{2}\mu-1} + r^{-\frac{1}{2}\mu+1} \quad (1)$$

form a basis of the algebraic integers of the field H_m . For by § 170, (7),

$$\begin{aligned} \omega = & x_0 + x_1 r + x_2 r^2 + \dots + x_{\frac{1}{2}\mu-1} r^{\frac{1}{2}\mu-1} + x_{\frac{1}{2}\mu} r^{\frac{1}{2}\mu} \\ & + x_{-1} r^{-1} + x_{-2} r^{-2} + \dots + x_{-\frac{1}{2}\mu+1} r^{-\frac{1}{2}\mu+1} \end{aligned} \quad (2)$$

is an algebraic integer of the field Ω_m if and only if the coefficients $x_0, x_1, \dots$ are rational integers.

The condition that this number should belong to the field H_m is obtained by carrying out the substitution (r, r^{-1}) and setting the difference equal to zero. If one then also divides by $r - r^{-1}$, one obtains

$$\begin{aligned} 0 = & (x_1 - x_{-1}) + (x_2 - x_{-2})(r + r^{-1}) + \dots \\ & + (x_{\frac{1}{2}\mu-1} - x_{-\frac{1}{2}\mu+1}) \frac{r^{\frac{1}{2}\mu-1} - r^{-\frac{1}{2}\mu+1}}{r - r^{-1}} + x_{\frac{1}{2}\mu} \frac{r^{\frac{1}{2}\mu} - r^{-\frac{1}{2}\mu}}{r - r^{-1}}. \end{aligned}$$

The divisions can be carried out here; and if one then multiplies by $r^{\frac{1}{2}\mu-1}$, there results for r a rational equation of degree lower than the μ th, which can be satisfied only if all its coefficients vanish. This leads to the equations

$$x_{\frac{1}{2}\mu} = 0, \quad x_{\frac{1}{2}\mu-1} = x_{-\frac{1}{2}\mu+1}, \dots, \quad x_1 = x_{-1},$$

and hence every algebraic integer ω of the field H_m is contained in the form

$$\omega = x_0 + x_1(r + r^{-1}) + x_2(r^2 + r^{-2}) + \dots + x_{\frac{1}{2}\mu-1}(r^{\frac{1}{2}\mu-1} + r^{-\frac{1}{2}\mu+1}).$$

Instead of the basis (1), if one sets

$$\rho = r + r^{-1},$$

one may also choose the powers of ρ :

$$1, \quad \rho, \quad \rho^2, \dots, \quad \rho^{\frac{1}{2}\mu-1} \quad (3)$$

as a basis of the algebraic integers of H_m . For the quantities (3) can be expressed integrally in terms of the quantities (1), and conversely.

The number ρ is the root of an irreducible equation of degree $\frac{1}{2}\mu$, which we shall denote by $\psi(\rho) = 0$. If $f(r) = 0$ is the equation for r (§ 169), then the identical relation

$$f(x) = x^{\frac{1}{2}\mu} \psi\left(x + \frac{1}{x}\right) \quad (4)$$

holds; differentiating gives

$$f'(r) = r^{\frac{1}{2}\mu}\psi'(\rho)(1 - r^{-2}). \quad (5)$$

From this we can form the fundamental number Δ_1 of the field H_m , which, since H_m contains only real numbers, must be positive. If we denote by N_1 the norm with respect to H_m , this fundamental number is

$$\Delta_1 = \pm N_1\psi'(\rho).$$

But if N is the norm formed with respect to Ω_m , then

$$N\psi'(\rho) = [N_1\psi'(\rho)]^2 = \Delta_1^2,$$

and therefore formula (5), with regard to § 169, gives

$$\Delta = N(1 - r^{-2})\Delta_1^2.$$

By § 169, (7) and (10),

$$\begin{aligned} N(1 - r^{-2}) &= q && \text{for odd } q, \\ &= 4 && \text{for } q = 2, \\ \pm\Delta &= q\Delta_1^2 && \text{for odd } q, \\ &= 4\Delta_1^2 && \text{for } q = 2. \end{aligned} \quad (6)$$

If one substitutes therein

$$\Delta = \pm q^{q^{\kappa-1}[\kappa(q-1)-1]},$$

it follows that

$$\begin{aligned} \Delta_1 &= q^{q^{\kappa-1}\frac{q-1}{2} - \frac{q^{\kappa-1}+1}{2}} && \text{for odd } q, \\ &= 2^{(\kappa-1)2^{\kappa-2}-1} && \text{for } q = 2, \end{aligned} \quad (7)$$

so that Δ_1 is always a power of q .

§ 177. The prime ideals in the field H_m .

We now have to examine the question of the relation between the prime ideals of the field H_m and those of the field Ω_m .

If $\mathfrak{P}$ denotes any prime ideal in H_m of degree f_1 , then $\mathfrak{P}$ must be a divisor of a natural prime number p , and therefore (by § 171) $\mathfrak{P}$ can be divisible by the second or a higher power of a prime factor $\mathfrak{p}$ in Ω_m only if $p = q$.

If, however, $\mathfrak{P}$ is a divisor of q , then it must be a power of $\sigma = 1 - r$. Put, for instance,

$$\mathfrak{P} = \sigma^\lambda.$$

Taking the norm of this with respect to Ω_m , which is the square of the norm with respect to H_m , one obtains (by § 169)

$$q^{2f_1} = q^\lambda,$$

and hence $f_1 = \frac{1}{2}\lambda$. Since f_1 is an integer, λ must be at least equal to 2. But λ also cannot be greater than 2, since σ^2 is associated with the algebraic integer $(1 - r)(1 - r^{-1})$ contained in H_m , and therefore σ^2 must be divisible by $\mathfrak{P}$. Consequently $f_1 = 1$, and we obtain the theorem:

1. The prime number q is the $\frac{1}{2}\mu$ th power of a prime number of first degree existing in H_m .

Now let p be different from q , and let $\mathfrak{p}$ be a prime factor of $\mathfrak{P}$ in the field Ω_m , of degree f , which is carried by the substitution (r, r^{-1}) into $\mathfrak{p}'$. Then $\mathfrak{P}$ is divisible both by $\mathfrak{p}$ and by $\mathfrak{p}'$; on the other hand, $\mathfrak{p}\mathfrak{p}'$ is contained in H_m and is therefore divisible by $\mathfrak{P}$. Thus we have to distinguish whether $\mathfrak{p}$ and $\mathfrak{p}'$ are distinct or not.

If $\mathfrak{p}$ is distinct from $\mathfrak{p}'$, then $\mathfrak{P}$ is divisible by $\mathfrak{p}\mathfrak{p}'$, and it follows that $\mathfrak{P} = \mathfrak{p}\mathfrak{p}'$; taking norms gives $f_1 = f$.

If, however, $\mathfrak{p} = \mathfrak{p}'$, then $\mathfrak{P} = \mathfrak{p}$, and taking norms gives

$$2f_1 = f.$$

Which of these two cases occurs depends, therefore, on whether the prime ideal $\mathfrak{p}$ admits the substitution (r, r^{-1}) or not, that is, whether (r, r^{-1}) is contained in the group to which the prime ideal belongs or not. By § 172 this difference is determined by whether there is some exponent h for which the congruence

$$p^h \equiv -1 \pmod{m}$$

is satisfied, or whether there is no such exponent.

We summarize the result as follows:

2. The natural prime numbers p different from q split into two kinds, p_1, p_2 .

The first kind p_1 is characterized by the condition that for some exponent h

$$p_1^h \equiv -1 \pmod{m},$$

and these prime numbers, if they belong to the exponent f modulo m and if $\mu = ef$, split in the field H_m into e prime factors of degree $\frac{1}{2}f$. Their prime factors are not further decomposable in Ω_m either.

The prime numbers of the second kind p_2 are determined by the condition that none of their powers is congruent to -1 modulo m ; they split in the field H_m into $\frac{1}{2}e$ prime factors of degree f . Their prime factors are still decomposable in Ω_m into two factors each.

For prime numbers of the first kind, f must certainly be even. If m is odd, it is also sufficient, in order that p be a prime number of the first kind, that it belong to an even exponent; for then

$$(p^{\frac{1}{2}f} - 1)(p^{\frac{1}{2}f} + 1) \equiv 0 \pmod{m}.$$

The first of these factors, $p^{\frac{1}{2}f} - 1$, is not divisible by m , since otherwise p would belong to the exponent $\frac{1}{2}f$. Consequently $p^{\frac{1}{2}f} + 1$ must be divisible by q . But here both factors $p^{\frac{1}{2}f} - 1$ and $p^{\frac{1}{2}f} + 1$ cannot be divisible by q , since otherwise their difference, which is equal to 2, would also be divisible by q . Hence $p^{\frac{1}{2}f} + 1$ is divisible by m .

If, however, m is a power of 2, then the congruence

$$p^h \equiv -1 \pmod{m}$$

is possible only if $p \equiv -1 \pmod{m}$. For every even power of an odd number is $\equiv +1 \pmod{8}$ and therefore cannot be $\equiv -1 \pmod{m}$. But if h is odd, then

$$\frac{p^h + 1}{p + 1} = p^{h-1} - p^{h-2} + \dots + 1$$

is itself odd, and consequently $p^h + 1$ is divisible by no higher power of 2 than $p + 1$. Thus we may add to Theorem 2 the following more precise determination:

3. If m is odd, then the prime numbers p_1 belong to an even exponent, and the prime numbers p_2 to an odd exponent.

If m is even, then the prime numbers of the form $km - 1$ belong to the first kind, and all other odd prime numbers to the second kind.

§ 178. The units of the field H_m .

The units belonging to the field H_m are of special importance for the applications. The units of the field Ω_m may also be reduced to them, according to the following theorem:

1. Every unit of the field Ω_m is the product of a root of unity and a real unit of the field H_m .

Denote any unit of the field Ω_m by $\mathcal{E}(r)$. Then $\mathcal{E}(r^{-1})$ is the conjugate-imaginary value, which likewise represents a unit, and the quotient $\mathcal{E}(r) : \mathcal{E}(r^{-1})$ is again a unit whose absolute value

$$\sqrt{\frac{\mathcal{E}(r)}{\mathcal{E}(r^{-1})} \frac{\mathcal{E}(r^{-1})}{\mathcal{E}(r)}} \quad (1)$$

is equal to 1. Consequently this quotient is a root of unity and hence is $= \pm r^h$, where h is some integral exponent (§ 175, Theorems 1 and 2). We therefore have

$$\mathcal{E}(r) = \pm r^h \mathcal{E}(r^{-1}). \quad (2)$$

In the rest of the proof a small distinction has now to be made according as m is odd or even.

First suppose that m is odd. Then in (2) we may take the exponent h to be even, since, if necessary, we may replace it by $h + m$. Moreover, in formula (2) only the upper sign is admissible. For if the lower sign stood there, it would follow that

$$\mathcal{E}(r) \equiv \mathcal{E}(1) \equiv -\mathcal{E}(1), \quad 2\mathcal{E}(1) \equiv 0 \pmod{1-r}.$$

But by § 169, $1 - r$ is a divisor of q , and so is relatively prime to 2; therefore $\mathcal{E}(1)$, and consequently also $\mathcal{E}(r)$, is divisible by $(1 - r)$. This contradicts the supposition that $\mathcal{E}(r)$ is to be a unit.

Consequently (2) gives

$$r^{-\frac{1}{2}h} \mathcal{E}(r) = r^{\frac{1}{2}h} \mathcal{E}(r^{-1}),$$

whence it follows that $r^{-\frac{1}{2}h} \mathcal{E}(r)$ is a real unit. Denoting it by $e(r)$, we therefore have

$$\mathcal{E}(r) = r^{\frac{1}{2}h} e(r), \quad (3)$$

which proves Theorem 1 in this case.

If, secondly, m is a power of 2, then $\mu = \frac{1}{2}m$ and $r^\mu = -1$; and along with r , $-r$ is also an m th root of unity. If necessary replacing h by $h + \frac{1}{2}m$, we may assume the upper sign in (2) and write

$$\mathcal{E}(r) = r^h \mathcal{E}(r^{-1}). \quad (4)$$

The point is now to prove that h must be even.

If we represent $\mathcal{E}(r)$ in the basis $1, r, r^2, \dots, r^{\mu-1}$, then

$$\mathcal{E}(r) = \sum_{s=0}^{\mu-1} x_s r^s, \quad (5)$$

where the x_s are rational integers. But by (4) we then have

$$\sum_{s=0}^{\mu-1} x_s r^s = \sum_{s=1}^{\mu-1} x_s r^{h-s},$$

and from this it follows that $x_s = \pm x_{s'}$ whenever $s + s' \equiv h \pmod{\mu}$. If h is odd, then of two numbers so connected one is even and the other odd (since μ is a power of 2). Thus the expression (5) gives for $\mathcal{E}(r)$ an aggregate of terms of the form

$$x_s(r^s \pm r^{s'}).$$

But

$$r^s \pm r^{s'} = r^s(1 \pm r^{s'-s}) = r^s(1 \mp r^{\mu+s'-s})$$

is always divisible by $1 - r$, and from this it would follow that $\mathcal{E}(r)$ is divisible by $1 - r$, which is not a unit, contrary to the concept of a unit. Therefore the assumption that the exponent h in formula (4) is odd is inadmissible, and, as above, we may put

$$\mathcal{E}(r) = r^{\frac{1}{2}h} e(r),$$

where $e(r)$ is a real unit. Theorem 1 is thereby proved.

We shall single out a system of real units. By § 169 the μ quantities $1 - r^n$ are all associated with one another. Therefore the quotient

$$\frac{1 - r^n}{1 - r^{n'}},$$

where n, n' are two numbers relatively prime to m , is a unit. From this, however, setting

$$r = e^{\frac{2\pi i}{m}},$$

one derives the real unit

$$\frac{\sin \frac{n\pi}{m}}{\sin \frac{n'\pi}{m}} = \frac{r^{\frac{n}{2}} - r^{-\frac{n}{2}}}{r^{\frac{n'}{2}} - r^{-\frac{n'}{2}}} = r^{\frac{n'-n}{2}} \frac{1 - r^n}{1 - r^{n'}}. \quad (6)$$

If m is even, one may set $n' = n + \frac{m}{2}$, and obtains in this case the real units

$$\tau_n = \tan \frac{n\pi}{m}, \quad (7)$$

where n may denote any odd number.

Replacing n by $\frac{1}{2}m - n$ carries τ_n into its reciprocal value. If n runs through the sequence of numbers

$$1, 3, 5, \dots, \frac{m}{4} - 1,$$

then all the values τ_n obtained are positive proper fractions.

[source p. 648]

Twenty-second Section. Abelian fields and cyclotomic fields.

§ 179. Simple Abelian fields.

We now turn to one of the most interesting applications of the theory of algebraic numbers.

The matter is the proof of the general theorem:¹

I. All Abelian number fields in the absolute domain of rationality are cyclotomic fields.

Once we have proved this theorem, the investigations of the third section on cyclotomic fields acquire an increased interest, because it is thereby shown that on the path indicated there one finds not only all cyclotomic fields, but all Abelian fields whatever.

Let

$$\Omega = R(x)$$

be an Abelian field of degree m , that is, a field consisting of all rational functions of a root x of an irreducible Abelian equation of degree m , when the field R of rational numbers is regarded as the domain of rationality. The Galois group $\mathfrak{G}$ of this equation, which is therefore an Abelian group and has the same degree m as the field Ω , is also the group of the field Ω .

If x_1 is a non-primitive number from Ω , then $\Omega_1 = R(x_1)$ is likewise an Abelian field, which we call a proper divisor of Ω , and whose degree is a proper divisor of the degree of Ω . For if $\mathfrak{A}$ is the group to which the number x_1 belongs, then $\mathfrak{G} \mid \mathfrak{A}$ (§ 4 of this volume) is the group of the field Ω_1 , and this is, together with $\mathfrak{G}$, an Abelian group. A divisor of Ω which has the same degree as Ω is identical with Ω (cf. Vol. I, §§ 144, 149, 156).

[source p. 649]

If there are now two non-primitive numbers x_1, x_2 in Ω of such a kind that

$$\Omega = R(x_1, x_2)$$

can be put, then the fields $\Omega_1 = R(x_1)$, $\Omega_2 = R(x_2)$ are proper divisors of Ω , and Ω is called *composed* from the two fields Ω_1, Ω_2 .

If the field Ω permits no such representation, it is called *simple*.

Thus, if it is assumed as proved that Ω_1, Ω_2 are cyclotomic fields, that is, that all their numbers are representable rationally by roots of unity, then the same follows for Ω .

If one of the fields Ω_1, Ω_2 is decomposable, then we can repeat the decomposition, and, since the degrees are decreasing positive integers, we must finally arrive at simple fields.

Theorem I therefore needs to be proved only for simple Abelian fields.

It is, however, to be recalled that the decomposability of a field Ω by no means already follows from the existence of a divisor, and that in a decomposable field the decomposition into simple fields can occur in quite different ways, so that for fields the laws of divisibility do not hold as they do in the domain of numbers.

A criterion for the simplicity of a field can be derived from its group.

¹Kronecker first stated this theorem, but did not publish a complete proof of it. The first proof was made known by the author of the present work in vol. 8 of *Acta Mathematica* (1886). Very recently a second proof by Hilbert has been published, which is based on the theorems developed in the nineteenth section (*Nachrichten d. Gesellschaft d. Wissenschaften zu Göttingen*, 1896, issue 1).

If the group $\mathfrak{G}$ has two proper divisors $\mathfrak{A}$ and $\mathfrak{B}$ of such a kind that every element can be composed in one and only one way from an element of $\mathfrak{A}$ and an element of $\mathfrak{B}$, then we call the group $\mathfrak{G}$ decomposable into the two components $\mathfrak{A}, \mathfrak{B}$; thus $\mathfrak{G}$ is decomposable when, in the product formed according to the composition of parts (§ 4),

$$\mathfrak{G} = \mathfrak{A}\mathfrak{B}, \quad (1)$$

each element of $\mathfrak{G}$ appears once and only once. The degree of $\mathfrak{G}$ is then equal to the product of the degrees of $\mathfrak{A}$ and $\mathfrak{B}$.

If there are no such divisors, the group $\mathfrak{G}$ is called indecomposable.

[source p. 650]

Then we can prove the theorem:

1. If the group of an Abelian field is decomposable, then the field too is composed.

To prove this, assume that the group $\mathfrak{G}$ of the field Ω is decomposable according to formula (1), and let m, a, b be the degrees of $\mathfrak{G}, \mathfrak{A}, \mathfrak{B}$. We take a number ξ of the field Ω belonging to the group $\mathfrak{B}$, which is therefore transformed by the substitutions of $\mathfrak{G}$ into a different conjugate values, and likewise a b -valued number η belonging to $\mathfrak{A}$. We can then form a number

$$x = \alpha\xi + \beta\eta$$

with rational numerical coefficients α, β , which has m different conjugate values, and then $\Omega = R(x)$. Thus Ω is composed from $R(\xi)$ and $R(\eta)$ (Vol. I, § 143).

If we now recall the representation of an Abelian group by a basis derived in §§ 9, 10 of this volume, then repeated application of what has just been proved gives:

2. Every Abelian field Ω can be composed from such fields whose degree is a power of a prime number and whose group is cyclic. The degrees of these composing fields are the invariants of the group of Ω .

These theorems are supplemented by the following:

3. An Abelian field of prime-power degree with cyclic group is simple.

If the group $\mathfrak{G}$ is cyclic, then its elements can be represented by the powers of a basis, G^γ , when γ runs through a complete system of residues modulo m . We obtain all divisors $\mathfrak{A}$ of $\mathfrak{G}$ represented by

$$\mathfrak{A} = G^{b\gamma},$$

when b is a divisor of m and γ runs through a complete system of residues modulo $a = m : b$.

If

$$\mathfrak{A}' = G^{b'\gamma}$$

is a second divisor of $\mathfrak{G}$ of degree a' and if

$$b' \leq b, \quad a' \geq a,$$

then, when we now assume that m and hence also a, b, a', b' are powers of a prime number q , b' is a divisor of b , and consequently $\mathfrak{A}$ is a divisor of $\mathfrak{A}'$.

[source p. 651]

If therefore ξ is a number of the field Ω which belongs to the group $\mathfrak{A}$, then ξ is a primitive number of the field $R(\xi)$ of degree b , and in this field every number ξ' is also contained which admits the substitutions of the group $\mathfrak{A}'$. Thus if $\mathfrak{A}, \mathfrak{B}$ and $\mathfrak{A}', \mathfrak{B}'$ are proper divisors of $\mathfrak{G}$, then $R(\xi)$ is different from Ω , and $R(\xi)$ is not enlarged by the adjunction of ξ' . Hence Ω also cannot be composed from $R(\xi)$ and $R(\xi')$, whereby 3 is proved.

§ 180. The resolvents.

According to what has now been proved, in the further considerations, whose aim is the proof of Theorem I, we may restrict ourselves to such Abelian fields whose degree is a prime power and whose group is cyclic. For this reason it also sufficed, for the intended application, that in the preceding section we restricted ourselves to the consideration of the cyclotomic fields Ω_m in which m was a prime power. We retain as much as possible of the notation used there and put

$$m = q^\kappa, \quad (1)$$

where q is a prime number and κ a positive exponent; and in the case where $q = 2$ we assume $\kappa > 1$. By n we denote any number not divisible by q . Let r be a primitive m -th root of unity, and Ω_m the field of rational functions of r .

Let now $\Omega = R(x)$ be a simple Abelian field of degree m , so that x is a root of a rational irreducible cyclic equation of degree m . We denote the roots of this equation, in a suitable order, by

$$x_0, x_1, x_2, \dots, x_{m-1}, \quad (2)$$

and assume $x_m = x_0$, $x_{m+1} = x_1$, and so on. Then

$$x_1 = \Phi(x_0), \quad x_2 = \Phi(x_1), \dots, \quad x_0 = \Phi(x_{m-1}),$$

where Φ denotes an integral function with rational coefficients, and the group of the field Ω consists of the powers of the substitution (x_0, x_1) , or of the cyclic permutations

[source p. 652]

of the roots (2). Every cyclic function of these quantities is a rational number. What we have to prove is that the x can be expressed rationally by roots of unity. If we adjoin to the field Ω the m -th root of unity r , there arises a field $R(x, r)$ which contains the two fields $\Omega = R(x)$ and $R(r) = \Omega_m$ as divisors.

If a number $\omega = \Phi(x, r)$ of the field $R(x, r)$ admits all substitutions (x_0, x_1) , (x_0, x_2) , $\dots$, (x_0, x_{m-1}) , then it is contained in the field Ω_m ; for then

$$m\omega = \Phi(x_0, r) + \Phi(x_1, r) + \dots + \Phi(x_{m-1}, r),$$

and the coefficients of the powers of r in this expression are not only cyclic but even symmetric functions of the roots (2), hence rational numbers. This remains true even in the case where the equation for x is reduced by adjunction of r .

We apply this to the resolvents

$$(r, x_0) = x_0 + rx_1 + r^2x_2 + \dots + r^{m-1}x_{m-1}, \quad (3)$$

and, more generally, when s denotes an arbitrary integral exponent,

$$(r^s, x_0) = x_0 + r^sx_1 + r^{2s}x_2 + \dots + r^{(m-1)s}x_{m-1}. \quad (4)$$

By these resolvents x_0 itself can again be rationally expressed, namely

$$mx_0 = \sum_{s=0}^{m-1} (r^s, x_0), \quad (5)$$

and if therefore we can show that all these resolvents (r^s, x_0) are cyclotomic numbers, then we have reached our goal. If we put

$$(r^s, x_\lambda) = x_\lambda + r^sx_{\lambda+1} + r^{2s}x_{\lambda+2} + \dots + r^{(m-1)s}x_{\lambda+m-1}, \quad (6)$$

then (r^s, x_λ) arises from (r^s, x_0) by the substitution (x_0, x_λ) , and the fundamental relation holds:

$$r^{\lambda s}(r^s, x_\lambda) = (r^s, x_0). \quad (7)$$

All these resolvents are numbers of the field $R(x, r)$; we consider preferably the following two combinations:

$$(r^s, x_0)^m = \omega_s, \quad (8)$$

$$(r^s, x_0)^{-1}(r, x_0)^s = \alpha. \quad (9)$$

[source p. 653]

These two numbers admit, as follows immediately from (7), the substitutions (x_0, x_λ) and are therefore numbers of the field Ω_m .

If n is not divisible by q , then, because of the irreducibility of the cyclotomic equation, none of the resolvents (r^n, x_0) can vanish unless they all vanish; for the equation

$$(r, x_0)^m = 0$$

would be an equation for r with rational coefficients, admitting all substitutions (r, r^n) .

Likewise, none of the resolvents (r^s, x_0) can vanish unless the whole family vanishes in which s has a fixed greatest common divisor with m .

If all (r, x_0) vanish, formula (9) is not applicable. Then, however, we can directly consider the resolvents (r^q, x_0) , which belong to a field of degree mq^{-1} arising from

$$y_0 = x_0 + x_q + x_{2q} + \cdots + x_{m-q},$$

and if these quantities too vanish, we pass to the resolvents (r^{q^2}, x_0) , and so on.

If, however, (r, x_0) does not vanish, then by formula (9) all (r^s, x_0) can be represented rationally by (r, x_0) and by cyclotomic numbers.

Consequently it is sufficient for our purpose if we can prove that the non-vanishing resolvents (r, x_0) are cyclotomic numbers.

Let n run through a complete system of incongruent numbers not divisible by q . Then

$$\sum n \equiv 0 \pmod{m},$$

for the numbers n split into pairs which complement one another to m . From (7) it follows that the product

$$\prod_n (r^n, x_0) = a \quad (10)$$

is a number in Ω_m . But since this number moreover is not changed when any of the substitutions (r, r^n) is performed in it, a is a rational number.

[source p. 654]

§ 181. Preparation for the proof.

After the developments of the preceding paragraph, for the proof of the great Theorem I it remains only to show that the number contained in Ω_m ,

$$\omega = (r, x_0)^m,$$

is the m -th power of a cyclotomic number.

1. The essential property of the numbers ω_n conjugate to ω , on which the proof rests, is that

$$\omega_n^{-1} \omega_1^n = \alpha^m \quad (1)$$

is the m -th power of a number in Ω_m .

Here n may be any number not divisible by q . If we form, of the right- and left-hand sides of (1), the norms in the field Ω_m , then, denoting by a the norm of the number α , which is a rational number, and observing that every ω_n has the same norm as ω , we obtain

$$N(\omega)^{n-1} = a^m.$$

If now m is odd, then we can put $n = 2$ in this, and obtain as a consequence of 1:

2. The norm of ω is the m -th power of a rational number,

$$N(\omega) = a^m. \quad (2)$$

If m is a power of 2, this inference is no longer permissible. It then follows from 1 only that $N(\omega)^2$ is the m -th power of a rational number. Nevertheless, as is seen from (10) of the preceding paragraph, the numbers ω must, also in this case, still satisfy condition 2; but this condition is then no longer contained automatically in 1, and is added as a new property.

What we have to prove is that from 1 and 2 it follows that ω itself is the m -th power of a cyclotomic number, even if in a higher field.

For this it is necessary to investigate the decomposition of the numbers ω into their prime factors in the field Ω_m .

The prime number q is, as was proved in § 169, associated with the $\varphi(m)$ -th power of a prime number $\sigma = 1 - r$ existing in Ω_m .

[source p. 655]

If therefore σ^k is the power of σ contained in ω , we put

$$\omega = \sigma^k \omega',$$

where ω' is relatively prime to the prime number q . Formation of the norm gives, by 2,

$$a^m = q^k b,$$

where b is a rational number which, in lowest terms, contains the prime number q neither in the numerator nor in the denominator. From this it follows that k must be divisible by m , and hence the first result:

3. The exponent of the prime number σ contained in ω is always divisible by m .

Let now p be a prime number different from q , and let $\mathfrak{p}_\xi$ be a prime factor of p in the field Ω_m . Let ξ run through the sequence of numbers $\xi_1, \xi_2, \dots, \xi_e$ determined more precisely in § 172 in the different cases, so that

$$p = \mathfrak{p}_{\xi_1} \mathfrak{p}_{\xi_2} \cdots \mathfrak{p}_{\xi_e}, \quad (3)$$

and assume that $\mathfrak{p}_\xi^{k_\xi}$ is the power of $\mathfrak{p}_\xi$ contained in ω . The power of $\mathfrak{p}_{n\xi}$ contained in ω_n is then $\mathfrak{p}_{n\xi}^{k_\xi}$; and if we determine n' from the congruence

$$nn' \equiv 1 \pmod{m}, \quad (4)$$

then $\mathfrak{p}_\xi^{k_{n'\xi}}$ is the power of $\mathfrak{p}_\xi$ contained in ω_n . Accordingly, in

$$\omega_n^{-1} \omega_1^n \quad (5)$$

the power $\mathfrak{p}_\xi^{nk_\xi - k_{n'\xi}}$ is contained, and by 1 the exponent of this power must be divisible by m .

Thus for every n not divisible by q we obtain the congruence

$$nk_\xi \equiv k_{n'\xi} \pmod{m}, \quad (6)$$

in which we can now also replace ξ by $n\xi$, whereby

$$nk_{n\xi} \equiv k_\xi \pmod{m} \quad (7)$$

is obtained. If in this we take $\xi = 1$ and put ξ, ξ' for n, n' and k for k_1 , then from (7) follows

$$k_\xi \equiv \xi' k \pmod{m}, \quad (8)$$

where k now has the same value for all ξ , and ξ, ξ' are connected by the congruence

$$\xi \xi' \equiv 1 \pmod{m}.$$

[source p. 656]

If we take $n = p$, then $\mathfrak{p}_{p\xi} = \mathfrak{p}_\xi$ (§ 172), and so also $k_{p\xi} = k_\xi$; and from (7) and (8) follows

$$k(p-1) \equiv 0 \pmod{m}. \quad (9)$$

Important consequences now follow from this. First, if $p-1$ is not divisible by q , then it follows that k must be divisible by m , and we therefore have the theorem:

4. The prime factors of a prime number p which is not congruent to 1 modulo q are contained in ω only in such powers whose exponents are divisible by m .

If, in the case where m is a power of 2, $p-1$ is divisible by 2 but not by 4, that is, if $p \equiv -1 \pmod{4}$, then it follows from (9) only that k is divisible by $\frac{1}{2}m$, not that k is divisible by m . The exponent of the power of $\mathfrak{p}_\xi$ contained in ω is, apart from multiples of m , equal to k ; and if k is divisible by m , everything is as in case 4. But if k is divisible only by $\frac{1}{2}m$, not by m , so that

$$k \equiv \frac{1}{2}m \pmod{m},$$

then $k - \frac{1}{2}m$ is divisible by m , and with reference to the decomposition (3) we can formulate the theorem as follows:

5. If m is a power of 2 and p is a prime number congruent to -1 modulo 4, then the exponent h can be so determined that all prime factors of p in

$$\omega \sqrt{p}^{-hm}$$

are contained only in such powers whose exponent is divisible by m .

Finally let $p-1$ be divisible by q , or, if m is even, by 4. Let m_1 be the greatest common divisor of m and $p-1$, and let

$$m = m_1 m_2.$$

Then by (9) k must be divisible by m_2 ; and if we put

$$k = h m_2,$$

then by (8) the product contained in ω is

$$\left(\prod_{\xi} \mathfrak{p}_{\xi}^{\xi'} \right)^{hm_2}. \quad (10)$$

In addition, the prime factors $\mathfrak{p}_{\xi}$ occur in ω only in such powers whose exponents are divisible by m .

[source p. 657]

In this case ξ , by § 172, III, runs through a complete system of residues not divisible by q modulo m_1 , and in (10) we may understand by ξ' the least positive solution of the congruence

$$\xi \xi' \equiv 1 \pmod{m_1},$$

because, if we change the exponents ξ' in (10) modulo m_1 , only m -th powers of the prime factors $\mathfrak{p}$ are added.

Then we can apply Kummer's theorem [§ 174, (16)] to the product (10), replacing m there by m_1 , and obtain, if we understand by $r_1 = r^{m_2}$ an m_1 -th root of unity and by the η certain periods of the p -th roots of unity,

$$\prod_{\xi} \mathfrak{p}_{\xi}^{\xi'} = (r_1, \eta)^{m_1},$$

and consequently

$$\left(\prod_{\xi} \mathfrak{p}_{\xi}^{\xi'} \right)^{hm_2} = (r_1, \eta)^{hm}. \quad (11)$$

Consequently we have the theorem:

6. If p is a prime number and $p - 1$ is divisible by q , or, when $q = 2$, by 4, then the positive exponent h can be so determined that the number contained in Ω_m

$$\omega(r_1, \eta)^{-hm}$$

contains the prime factors of p only in such powers whose exponent is divisible by m .

It is now also no longer necessary to distinguish Theorem 5 from Theorem 6; for for $m_1 = 2$, 6 passes into 5, because then $r_1 = -1$, and by Vol. I, § 171,

$$(r_1, \eta)^m = (-1, \eta)^m = \sqrt{p^m}$$

holds.²

If we collect the result of Theorems 3 to 6 in one formula, we obtain, when ε denotes a unit functional, φ any functional of the field Ω_m , and p runs through a sequence of prime numbers congruent to 1 modulo q , among which the same prime number p may occur several times,

$$\omega = (r, x_0)^m = \varepsilon \varphi^m \prod_p (r^{m_2}, \eta)^m. \quad (12)$$

[source p. 658]

²In the author's paper in vol. 8 of *Acta Mathematica* it was omitted to emphasize separately the case of Theorem 5.

The resolvents of the cyclotomy occurring in these formulae, (r^{m_2}, η) , are cyclotomic numbers in a higher field. But their m -th powers are numbers contained in the field Ω_m which are transformed by the substitution (r, r^n) into

$$(r^{nm_2}, \eta)^m.$$

Likewise the combinations

$$(r^{nm_2}, \eta)^{-1} (r^{m_2}, \eta)^n \quad (13)$$

are contained in the field Ω_m [§ 174, (9), (11)]. These resolvents are special cases of the general resolvent (r, x_0) . From (12), if we denote by φ_n, ε_n the functionals conjugate to φ and ε , we obtain

$$\omega_n = \varepsilon_n \varphi_n^m \prod_p (r^{m_2 n}, \eta)^m. \quad (14)$$

This formula teaches us the most important properties of the functionals φ_n and ε_n , first:

7. The functionals φ_n^m are associated with numbers of the field Ω_m , and therefore belong to the principal class (§ 152).

By 1, $\omega_n^{-1} \omega_1^n = \alpha^m$ is the m -th power of a number of Ω_m . Since the combinations (13) are also numbers in Ω_m , we can put

$$\prod_p (r^{nm_2}, \eta)^{-1} (r^{m_2}, \eta)^n = \frac{\alpha}{\beta},$$

and obtain from (14)

$$\varepsilon_n^{-1} \varepsilon_1^n (\varphi_n^{-1} \varphi_1^n)^m = \beta^m,$$

where β is a number in Ω_m .

It follows from this that the product

$$\beta \varphi_n \varphi_1^{-n} = \varepsilon$$

is a unit functional of the field Ω_m , and from this arise, for the functionals φ_n and the unit functionals ε_n , the following two theorems:

8. The conjugate functionals φ_n have the property that all products

$$\varphi_n \varphi_1^{-n}$$

belong to the principal class.

9. The unit functionals ε_n have the property that the products

$$\varepsilon_n \varepsilon_1^{-n} = \varepsilon^m$$

are m -th powers of unit functionals.

[source p. 659]

For the proof of the principal theorem I, everything now depends on the proof of the two propositions which are to be inferred from the properties of the functionals φ and ε :

A. The functionals φ_n belong to the principal class.

If we may presuppose this lemma, then φ_n is associated with a number α_n , and we can write formula (14), with a somewhat changed meaning of ε_n , in the form

$$\omega_n = \varepsilon_n \alpha_n^m \prod_p (r^{m_2 n}, \eta)^m. \quad (15)$$

In this formula, however, ε_n is a numerical unit of the field Ω_m satisfying theorem 9.

If, therefore, from theorem 9 we can still prove the second lemma:

B. A numerical unit ε_n of the field Ω_m , which satisfies theorem 9, is the m -th power of a unit in Ω_m , multiplied by a power of r ;

then in (15) we can extract the m -th root, and obtain, when by ρ a root of unity of possibly higher degree than m and by α a number of the field Ω_m is denoted,

$$(r, x_0) = \rho \alpha \prod_p (r^{m_2}, \eta), \quad (16)$$

where now on the right there stand nothing but cyclotomic numbers, and by this Theorem I is therefore proved.

[source p. 659]

§ 182. Proof of the First Auxiliary Lemma for an Odd m .

In the preceding paragraph we made the proof of Theorem I dependent on two auxiliary theorems, whose proof is now still to be sought.

We formulate the first of these auxiliary theorems as follows:

- a) If φ_n is a system of non-zero conjugate functionals of the field Ω_m , of which it is known that

$$\varphi_n^m \quad \text{and} \quad \varphi_n^{-1} \varphi_1^n$$

belong to the principal class, then the functionals φ themselves belong to the principal class.

The theorem will be proved if, for some power φ^k of φ whose exponent is not divisible by q , it can be proved that it belongs to the principal class. For if α, β are numbers in Ω_m and

$$\varphi^m = \varepsilon' \alpha, \quad \varphi^k = \varepsilon'' \beta,$$

then we can determine the rational integers x, y so that

$$mx + ky = 1,$$

and then

$$\varphi = \varphi^{mx} \varphi^{ky} = \varepsilon''' \alpha^x \beta^y;$$

thus, if $\varepsilon', \varepsilon'', \varepsilon'''$ are units, φ itself is associated with a number.

According to the present state of the matter, the proof for an even m is to be carried out by a quite different path from that for an odd m ; we therefore occupy ourselves first with the easier case of an odd m ³.

The assumptions made in a) about φ_n can, if α, β again denote arbitrary numbers of the field Ω_m and ε unit functionals, be expressed by the two formulas

$$\varphi_n^m = \varepsilon \beta, \quad (1)$$

$$\varphi_n = \varepsilon \alpha \varphi_1^n. \quad (2)$$

³Compare, moreover, the new proof by Hilbert, in which the difference between even and odd m recedes much more (p. 642).

We shall now solve the problem step by step.

First let, as before, σ be the prime factor contained in q , which is a number existing in Ω_m , and let σ^k be the power of σ contained in φ ; put

$$\varphi = \sigma^k \psi, \quad (3)$$

where ψ denotes a functional prime to q , which satisfies the same conditions (1), (2) as φ , and if one of the two belongs to the principal class, the same is true of the other. Thus theorem a) needs to be proved only under the assumption that φ is prime to q .

Now let p be a prime number different from q , whose decomposition into prime factors has the expression

$$p = \prod_{\xi} \mathfrak{p}_{\xi}, \quad (4)$$

where ξ runs through the number sequence more precisely determined in § 172. Let $\mathfrak{p}_{\xi}^{k_{\xi}}$ be the power of $\mathfrak{p}_{\xi}$ contained in φ , so that, if we set

$$\varphi = \chi \prod_{\xi} \mathfrak{p}_{\xi}^{k_{\xi}},$$

then χ is prime to p and is related to no other prime numbers (see the end of § 173) than φ itself. From this there results

$$\varphi_n = \chi_n \prod_{\xi} \mathfrak{p}_{n\xi}^{k_{\xi}}, \quad (5)$$

and in this we let n again run through the sequence of the numbers ξ . If we further set

$$\psi = \prod_{\xi} \chi_{\xi},$$

then ψ too is prime to p and related to no other prime numbers than φ , and from (5), using (4), we obtain

$$\prod_{\xi} \varphi_{\xi} = \psi p^{\sum k_{\xi}}, \quad (6)$$

and from this it follows that the functional ψ satisfies the same two conditions (1), (2) as φ .

It is further shown that, if the φ_{ξ} lie in the principal class, then ψ is also contained in the principal class.

But if $p - 1$ is not divisible by q , then we can also conclude the converse. For by (2), φ_{ξ} is equivalent to φ^{ξ} , and consequently

$$\prod_{\xi} \varphi_{\xi} \text{ is equivalent to } \varphi^{\sum \xi}, \quad \text{is equivalent to } \psi,$$

and if ψ is associated with a number, the same holds for $\varphi^{\sum \xi}$. By the theorem § 172, however, if $p - 1$ is not divisible by q , $\sum \xi$ is also not divisible by q , and consequently φ itself is contained in the principal class.

This consideration can now be repeated if ψ is still related to further prime numbers which are not congruent to 1 modulo q , and we arrive at the theorem:

Theorem a) is proved in general if it can be proved under the assumption that φ is related only to such prime numbers as are congruent to 1 modulo q .

If we make this assumption about the functionals φ_n , then we can apply Kummer's theorem in the form of § 174, 3.

In what follows we denote by ε unit functionals, whose more precise knowledge is immaterial, and set

$$\Phi_n = \prod_t \varphi_{nt'}^t = \varepsilon \vartheta_n^m, \quad (7)$$

where t runs through the residues of m not divisible by q which are smaller than m , and t' is determined by $tt' \equiv 1 \pmod{m}$. Then ϑ_n is a cyclotomic number in a higher field, whose m th power is contained in Ω_m , and which satisfies the second condition, namely that

$$\vartheta_n^{-1} \vartheta_1^n = \alpha \quad (8)$$

is a number of the field Ω_m .

We now introduce a frequently used sign, understanding by $E(x)$ the greatest integer contained in the real number x , and by (x) the remainder, which is always a proper fraction and, if x itself should be an integer, is to be put equal to zero. Then

$$x = E(x) + (x). \quad (9)$$

If x is an integer, then in this notation

$$m \left(\frac{x}{m} \right)$$

is also an integer, namely the least positive remainder of the division of x by m .

In formula (7), the index of the functionals φ is determined only modulo m . The exponent, however, is reduced to its least remainder modulo m . Thus if in (7) we replace the index t' by nt' , then t must be replaced by the remainder of the division of nt by m , that is, [by (9)] by

$$m \left(\frac{nt}{m} \right) = nt - mE \left(\frac{nt}{m} \right),$$

and we can set

$$\Phi_n = \prod_t \varphi_{t'}^{nt - mE(\frac{nt}{m})} = \varepsilon \vartheta_n^m,$$

whereas

$$\Phi_1 = \prod_t \varphi_{t'}^t = \varepsilon \vartheta_1^m.$$

Thus by (8)

$$\Phi_1^n \Phi_n^{-1} = \prod_t \varphi_{t'}^{mE(\frac{nt}{m})} = \varepsilon \alpha^m. \quad (10)$$

Accordingly the quotient

$$\left(\frac{\prod_t \varphi_{t'}^{E(\frac{nt}{m})}}{\alpha} \right)^m$$

is a unit functional, and since it is at the same time the m th power of a functional in Ω_m , the base of this power is also a unit, that is,

$$\prod_t \varphi_{t'}^{E(\frac{nt}{m})} = \varepsilon \alpha,$$

and this product belongs to the principal class. But by our assumption

$$\varphi_{t'} \text{ is equivalent to } \varphi^{t'},$$

and hence

$$\prod_t \varphi_{t'}^{E(\frac{nt}{m})} \text{ is equivalent to } \varphi^{\sum t' E(\frac{nt}{m})},$$

and this holds for every arbitrary n not divisible by q .

If therefore we can prove that n can be so chosen that also the sum

$$S_n = \sum_t t' E\left(\frac{nt}{m}\right)$$

is not divisible by q , then theorem a) is proved.

We first want to make a transformation of the sum S_n , by which the question is led back to the much simpler case in which m is a prime number.

If m is a higher than first power of q , we set

$$m = qm', \quad t = t_1 + qt_2, \quad \begin{array}{l} t_1 = 1, 2, \dots, q-1, \\ t_2 = 0, 1, \dots, m'-1. \end{array}$$

Here m' is still a power of q ; the summation letter t_1 runs through the number sequence $1, 2, \dots, q-1$, and t_2 through the number sequence $0, 1, 2, \dots, m'-1$. We determine t'_1 from the congruence

$$t_1 t'_1 \equiv 1 \pmod{q}$$

and obtain

$$t' \equiv t'_1 \pmod{q}.$$

Then

$$S_n = \sum_t t' E\left(\frac{tn}{m}\right) \equiv \sum_{t_1} t'_1 \sum_{t_2} E\left(\frac{nt_2}{m'} + \frac{nt_1}{m}\right) \pmod{q}. \quad (11)$$

We now assume $n < q$, hence also $nt_1 < m$, so that $nt_1 : m$ is a proper fraction. Then

$$\begin{array}{ll} \text{a)} & E\left(\frac{nt_2}{m'} + \frac{nt_1}{m}\right) = E\left(\frac{nt_2}{m'}\right), \\ \text{b)} & \text{or} = E\left(\frac{nt_2}{m'}\right) + 1, \end{array}$$

and case b) occurs only if between

$$\frac{nt_2}{m'} \quad \text{and} \quad \frac{nt_2}{m'} + \frac{nt_1}{m}$$

there lies an integer. This is the case only if the difference between $nt_2 : m'$ and the next greater integer is smaller than $nt_1 : m$, that is, if

$$E\left(\frac{nt_2}{m'}\right) + 1 - \frac{nt_2}{m'} < \frac{nt_1}{m}. \quad (12)$$

If in the expression on the left side of (12)

$$E\left(\frac{nt_2}{m'}\right) + 1 - \frac{nt_2}{m'} \quad (13)$$

we let t_2 run through all its values, we obtain positive rational fractions not exceeding unity, with denominator m' , no two of which are equal to one another. For if two of the values arising from (13) by substitution of t'_2, t''_2 for t_2 were equal, then $n(t'_2 - t''_2)/m'$ would have to

be an integer, which, since n is relatively prime to m' and $t'_2 - t''_2$ is smaller than m' , cannot be. Thus the expressions (13) run in some order through the numerical values

$$\frac{1}{m'}, \frac{2}{m'}, \dots, \frac{m' - 1}{m'}, 1, \quad (14)$$

and if a denotes an arbitrary positive fraction, then $E(m'a)$ is the number of numbers of the sequence (14) which are not greater than a . By (12), therefore, the number of values of t_2 for which case b) occurs is equal to $E(nt_1/q)$, and there results

$$\sum_{t_2} E\left(\frac{nt_2}{m'} + \frac{nt_1}{m}\right) = \sum_{t_2} E\left(\frac{nt_2}{m'}\right) + E\left(\frac{nt_1}{q}\right).$$

To form S_n , we multiply this equation by t'_1 and sum. This gives

$$S_n \equiv \sum_{t_1} t'_1 \sum_{t_2} E\left(\frac{nt_2}{m'}\right) + \sum_{t_1} t'_1 E\left(\frac{nt_1}{q}\right) \pmod{q}.$$

Here now $\sum t'_1 = \sum t_1 \equiv 0 \pmod{q}$, because the t_1 pairwise give the sum q , and it follows that

$$S_n \equiv \sum_{t_1} t'_1 E\left(\frac{nt_1}{q}\right) \pmod{q}. \quad (15)$$

The sum standing here on the right side is formed in the same way as S_n itself in the case where m is a prime number. For this case, however, there results

$$\begin{aligned} \frac{nt_1}{q} &= E\left(\frac{nt_1}{q}\right) + \left(\frac{nt_1}{q}\right), \\ \frac{(q-n)t_1}{q} &= E\left(\frac{(q-n)t_1}{q}\right) + \left(\frac{(q-n)t_1}{q}\right), \end{aligned}$$

and from this, by addition,

$$t_1 = E\left(\frac{nt_1}{q}\right) + E\left(\frac{(q-n)t_1}{q}\right) + \left(\frac{nt_1}{q}\right) + \left(\frac{(q-n)t_1}{q}\right),$$

and here the sum of the two positive proper fractions, which nevertheless must be an integer, can have only the value 1. It follows therefore that

$$E\left(\frac{nt_1}{q}\right) + E\left(\frac{(q-n)t_1}{q}\right) = t_1 - 1,$$

and from this, by multiplication with t'_1 and summation over all values of t_1 from 1 to $q-1$,

$$\sum_{t_1} t'_1 E\left(\frac{nt_1}{q}\right) + \sum_{t_1} t'_1 E\left(\frac{(q-n)t_1}{q}\right) \equiv -1 \pmod{q}.$$

Consequently both sums on the left side certainly cannot be divisible by q . Then expression (15) shows that of the two sums S_n and S_{q-n} one certainly is indivisible by q .

This, however, was to be proved.

[source p. 666]

§ 183. Proof of the Second Auxiliary Lemma for an Odd m .

The second lemma, which by § 181 remains to be proved, is the following:

2. If ε_n is a system of conjugate numerical units of the field Ω_m which satisfies the condition that

$$\varepsilon_n \varepsilon_1^{-n} = \mathcal{E}^m \quad (1)$$

is the m th power of a unit in Ω_m , then the ε_n themselves are m th powers of units in Ω_m , multiplied by a root of unity of the form r^{nk} .

This proof too is quite elementary for the case of an odd m . We therefore first consider this case here.

By § 178, 1., we can represent every unit of the field Ω_m in the form

$$\varepsilon = \pm r^k \mathcal{E}(r), \quad \varepsilon_n = \pm r^{nk} \mathcal{E}(r^n), \quad (2)$$

where $\mathcal{E}(r)$ is a real unit of the field Ω_m . Thus $\mathcal{E}(r)$ satisfies the condition

$$\mathcal{E}(r) = \mathcal{E}(r^{-1}), \quad (3)$$

from which, by (2), follows:

$$\varepsilon_1 \varepsilon_{-1} = \mathcal{E}(r)^2; \quad (4)$$

but from (1), putting $n = -1$, one finds

$$\varepsilon_1 \varepsilon_{-1} = \mathcal{E}^m,$$

that is, the left side of (4) is the m th power of a unit in Ω_m , and hence also

$$\mathcal{E}(r)^2 = \mathcal{E}^m \quad (5)$$

is the m th power of such a unit.

Since now m is odd, the rational integer x can be so determined that

$$2x + m = 1, \quad (6)$$

and from this follows

$$\mathcal{E}(r) = \mathcal{E}(r)^{2x} \mathcal{E}(r)^m.$$

Thus if, according to (5), we set

$$\mathcal{E}^x \mathcal{E}(r) = e(r),$$

then follows

$$\mathcal{E}(r) = e(r)^m. \quad (7)$$

By (7), however, Theorem 2 is proved, and at the same time, for an odd m , the whole Theorem I.

This conclusion too fails for an even m , because then equation (6) is no longer soluble.

[source p. 667]

§ 184. Preliminary Remarks on the Case of an Even m .

For the case where m is a power of 2, we take an entirely different path, about which some preliminary remarks may first find place here.

We have already seen in § 182 that the first lemma is proved if we can prove that some power of φ , whose exponent is not divisible by q , belongs to the principal class. In doing this, of the two properties of the functional φ only the first is used, namely that the m th power of φ is contained in the principal class.

Now by the general theorems in § 154 we know in every field an exponent h , namely the number of ideal classes, for which φ^h belongs to the principal class whenever φ is any functional in this field.

Thus if we can prove that the class number h in the field Ω_m , when m is a power of 2, is an odd number, then by this alone, without anything further, Lemma 1 is proved.

This is to be the aim of our considerations in the next sections.

With respect to the second lemma the matter is similar. Here, from the assumptions in § 183, 2., one can derive, exactly as above, formula § 183, (5), from which, however, it can only be concluded that $\mathcal{E}(r)$ is the $\frac{1}{2}m$ th power of a real unit $e(r)$.

Thus, assuming the first lemma as proved, formula § 181, (14) gives:

$$(r^n, x_0) = \rho_n \sqrt{e(r^n)} \alpha_n \prod_p (r^{m_2 n}, \eta), \quad (1)$$

where ρ_n is a root of unity of degree m^2 , and α_n is a number in Ω_m , and it would still have to be proved that $e(r^n)$ is the square of a unit in Ω_m .

Since $e(r)$ is a real unit, we have

$$e(r^n) = e(r^{-n}), \quad (2)$$

and we shall further stipulate that the sign of the root is to be so determined (which, by a suitable assumption about ρ_n , we can always assume) that

$$\sqrt{e(r^n)} = \sqrt{e(r^{-n})}. \quad (3)$$

By the assumption, the product $(r^n, x_0)(r^{-n}, x_0)$ is a number of the field Ω_m which is not changed by the substitution (r, r^{-1}) , that is, a real number. Further, by § 174, (5),

$$(r^{m_2 n}, \eta)(r^{-m_2 n}, \eta) = \pm p,$$

so it is likewise real. Similarly $\alpha_n \alpha_{-n}$ is real, and from (1) it follows that

$$\rho_n \rho_{-n} = \frac{(r^n, x_0)(r^{-n}, x_0)}{e(r^n) \alpha_n \alpha_{-n} \prod_p (r^{m_2 n}, \eta)(r^{-m_2 n}, \eta)}$$

is a real number, which, since it is a root of unity, can only be $= \pm 1$.

Now in the equation resulting from this for $n = 1$,

$$(r, x_0)(r^{-1}, x_0) = \pm e(r) \alpha_1 \alpha_{-1} \prod_p (\pm p),$$

we can carry out every substitution (r, r^n) , from which it follows that the sign of $\rho_n \rho_{-n}$ is independent of n , hence that

$$\rho_n \rho_{-n} = \rho_1 \rho_{-1} = \pm 1. \quad (4)$$

Now we further derive from (1):

$$\rho_n \rho_1^{-1} \sqrt{e(r^n)} \sqrt{e(r)}^{-n} = \frac{(r^n, x_0)(r, x_0)^{-n}}{\alpha_n \alpha_1^{-1} \prod_p (r^{m_2 n}, \eta)(r^{m_2}, \eta)^{-n}}, \quad (5)$$

and the right side is a number of the field Ω_m . But the left side shows that it is a unit, and therefore, if $\mathcal{E}(r)$ denotes a real unit, we can set

$$\rho_n \rho_1^{-1} \sqrt{e(r^n)} \sqrt{e(r)}^{-n} = r^k \mathcal{E}(r). \quad (6)$$

If we substitute this value into formula (5) and make the substitution (r, r^{-1}) , there results

$$\rho_{-n} \rho_{-1}^{-1} \sqrt{e(r^n)} \sqrt{e(r)}^{-n} = r^{-k} \mathcal{E}(r), \quad (7)$$

and by multiplication of (6) and (7), with regard to (4),

$$e(r^n) e(r)^{-n} = \mathcal{E}(r)^2. \quad (8)$$

In our considerations on the class number the following theorem will still result:

A unit $e(r)$ of the real field H_m which is positive with all its conjugates is the square of a unit in the field H_m .

Formula (8), however, shows that the unit $\pm e(r)$ has this property, and hence that $e(r)$ (since also $-1 = i^2$ is the square of a unit) is the square of a unit of the field Ω_m .

Thus the question has been reduced to the proof of the two mentioned theorems, which arises as the capstone of a long chain of considerations, but one extremely rich in interesting relations, to which the next sections are to be devoted.

[source p. 670]

Twenty-third Section. Class Number.

§ 185. Dirichlet's theorem on the units.

The investigations on the number of ideal classes in an algebraic field, which are now to give us the necessary supplements to the proofs of the preceding section, make use of the methods that Dirichlet introduced into number theory.⁴

They are no longer purely algebraic, insofar as transcendental functions and numbers, such as the natural logarithm and the number π , also occur in them. Use is also made of passages to the limit, as in integral calculus, from which we have already drawn a fine result in § 156.

The investigations with which we first have to deal are quite general, that is, they hold for every algebraic number field, and it offers no advantage for simplicity to restrict them to special fields, such as the cyclotomic fields.

⁴Dirichlet, *Recherches sur diverses applications de l'analyse infinitésimale à la théorie des nombres*. Crelle's Journal, vols. 19, 21. Dirichlet's Works, vol. I (1839, 1840). (The posthumous papers of Gauss, "De nexu inter multitudinem classium etc." Gauss' Works, vol. II, also belong here.) The application of these principles to cyclotomic numbers was first made by Kummer [Crelle's Journal, vols. 35, 40 (1847, 1850); Liouville's Journal, vol. 16 (1851)]. The general theory of units was likewise founded by Dirichlet [Dirichlet's Works, vol. I, pp. 622, 633, 639 (1840, 1842, 1846)]. The general theory of units and the determination of the class numbers was given by Dedekind: Dirichlet-Dedekind, *Vorlesungen über Zahlentheorie*, 4th ed., § 183 ff. Minkowski, *Geometrie der Zahlen*.

[source p. 671]

Thus let Ω be a field of the n th degree, and let

$$\Omega_1, \Omega_2, \dots, \Omega_n \quad (1)$$

be the conjugate fields. The irreducible equation of the n th degree whose roots determine these conjugate fields for us will in general have both real and imaginary roots; the latter, however, always occur pairwise, so that to every imaginary field $\Omega^{(i)}$ there belongs another whose numbers are conjugate imaginary to those of $\Omega^{(i)}$, and which therefore arises from $\Omega^{(i)}$ by the substitution $(i, -i)$.

We combine such imaginary pairs into one unit and denote by ν the number of real fields and of pairs of imaginary fields in the sequence (1). If all conjugate fields are real, then $n = \nu$; if they are all imaginary, then $n = 2\nu$.

Except in the case of the rational field, ν is equal to 1 only in the case of an imaginary quadratic field. Otherwise ν is always greater than 1.

The number of imaginary pairs is $n - \nu$, and the number of real fields is $2\nu - n$.

In the determinant by which, in § 145, the square root of the ground number, $\sqrt{\Delta}$, is defined, the substitution $(i, -i)$ interchanges two conjugate imaginary rows at a time, and $\sqrt{\Delta}$ therefore changes its sign precisely when this number is odd, and otherwise not. In the first case $\sqrt{\Delta}$ is imaginary, in the second real. Hence it follows that the ground number Δ of the field Ω is positive or negative according as $n - \nu$ is even or odd.

If from two conjugate imaginary fields we retain only one, there results the sequence of conjugate fields

$$\Omega_1, \Omega_2, \dots, \Omega_\nu,$$

to which we assign a system of signs

$$\delta_1, \delta_2, \dots, \delta_\nu$$

with the convention that $\delta_s = 1$ if Ω_s is real, and $= 2$ if Ω_s is imaginary, so that $\sum \delta_s = n$.

Let η denote a non-zero number of the field Ω and let

$$\eta_1, \eta_2, \dots, \eta_n$$

be the conjugate numbers. To this number system we assign another number system

$$\lambda_1, \lambda_2, \dots, \lambda_\nu,$$

[source p. 672]

which we define by the equation

$$\lambda_s = \delta_s \log |\eta_s|, \quad (2)$$

where $|\eta_s|$ denotes the absolute value of η_s , and $\log |\eta_s|$ denotes the real natural logarithm of the positive number $|\eta_s|$. If Ω_s is real and the sign $\pm$ is chosen so that $\pm \eta_s$ is positive, then

$$\lambda_s = \log(\pm \eta_s).$$

But if η_s, η'_s is an imaginary pair, then

$$\lambda_s = \log \eta_s \eta'_s,$$

and to η_s and η'_s belongs the same λ_s , so that the number of different λ_s is equal to ν . From this determination one further obtains

$$\lambda_1 + \lambda_2 + \cdots + \lambda_\nu = \log N_a(\eta), \quad (3)$$

where, as before, N_a denotes the absolute norm.

We shall call the numbers $\lambda_1, \lambda_2, \dots, \lambda_\nu$ the conjugate logarithms of the number η .

If η is an integer of the field Ω , then its absolute norm is a natural number, which is equal to 1 only when η is a unit; and from (3) it follows that:

1. The sum of the conjugate logarithms of an integer η is positive, and is equal to zero only when η is a unit.

Let $\omega_1, \omega_2, \dots, \omega_n$ denote a basis of the integers in Ω (a minimal basis of Ω , § 145), and let $\omega_{1,s}, \omega_{2,s}, \dots, \omega_{n,s}$, for $s = 1, 2, \dots, n$, denote the conjugate numbers. Then all integers of the field Ω_s can be represented in the form

$$\eta_s = x_1\omega_{1,s} + x_2\omega_{2,s} + \cdots + x_n\omega_{n,s}, \quad (4)$$

with rational integral coefficients $x_1, x_2, \dots, x_n$. The determinant of this system,

$$\sum \pm \omega_{1,1}\omega_{2,2} \cdots \omega_{n,n} = \sqrt{\Delta},$$

is the square root of the discriminant Δ of the field, and hence is always different from zero. Thus the system (4) can be solved for the unknowns $x_1, x_2, \dots, x_n$, and if we now require that the absolute values of the numbers η_s are not to exceed a certain bound, then these solutions give bounds for the rational integers x_i .

[source p. 673]

This simple remark gives us the following important theorem:

2. In an algebraic field Ω there is only a finite number of integers η of the property that the absolute values of the conjugate numbers η_s lie below a given bound.

We now make use of the general theorem § 155, (25), according to which, if $f(x_1, x_2, \dots, x_n)$ is any positive quadratic form in n variables with determinant D , the variables x_i can be determined as rational integers, not all zero, so that

$$f(x_1, x_2, \dots, x_n) < n \sqrt[n]{0.404D},$$

that is, becomes smaller than a number depending only on the determinant D .

We apply this theorem, as in § 156, to the form

$$f(x_1, x_2, \dots, x_n) = \frac{|\eta_1|^2}{c_1^2} + \frac{|\eta_2|^2}{c_2^2} + \cdots + \frac{|\eta_n|^2}{c_n^2},$$

where the η_s are the linear forms (4), and in the case of a real η_s

$$|\eta_s|^2 = (\eta_s)^2,$$

in the case of an imaginary pair η_s, η'_s

$$|\eta_s|^2 = |\eta'_s|^2 = \eta_s \eta'_s.$$

In the latter case $\eta_s \eta'_s$ is the sum of two real squares, and hence, if the c_s are taken to be real, f is a positive form. We further stipulate, for the hitherto arbitrary real quantities c_s , that if η_s, η'_s form a conjugate imaginary pair, then

$$c_s = c'_s \quad (5)$$

is to hold; besides this we subject the c_s to the condition

$$c_1 c_2 \cdots c_n = 1. \quad (6)$$

We form the determinant of the function f , as in § 156, by first regarding it as a form in the variables $\eta_1, \eta_2, \dots, \eta_n$, and the determinant of this form is, because of (6), equal to ± 1 . The determinant of f with respect to the variables x_i is therefore (Vol. I, § 56), apart from the sign, equal to the

[source p. 674]

square of the determinant of the linear forms η_s , that is, to the ground number of the field Ω .

It follows that one can find a number A , independent of the c_s and determined only by the field Ω , of such a kind that, whatever the c_s may be, $f(x_1, x_2, \dots, x_n)$ for integral non-zero x drops below A . Since f is a sum of positive summands, the same holds for each of these summands, and the following theorem is thereby proved:

3. If c_s is any system of real positive numbers satisfying the conditions (5), (6), then there is a real number A , determined by the field Ω alone, such that one can always determine an integer η of the field Ω which, together with its conjugate numbers η_s , satisfies the conditions

$$|\eta_s| < c_s A. \quad (7)$$

Incidentally it follows from this that in every algebraic field Ω , except the rational and the imaginary quadratic field, one can find non-zero integers whose absolute value falls below any given bound.

We shall transform the expression of theorem 3 somewhat by introducing the conjugate logarithms of η . Put $\log A = k$, so that k is a real number determined by Ω , and further

$$c_s = e^{\frac{\gamma_s u}{\delta_s}}, \quad (8)$$

where u may be any real quantity, and where the numbers γ_s satisfy, because of (5) and (6), the condition

$$\gamma_1 + \gamma_2 + \cdots + \gamma_\nu = \sum_{s=1}^{\nu} \gamma_s = 0. \quad (9)$$

Since, moreover,

$$|\eta_s| = e^{\frac{\lambda_s}{\delta_s}},$$

it follows from (7) and (8) that

$$\lambda_s - \gamma_s u < k \delta_s. \quad (10)$$

The sum of the ν expressions on the left side of this inequality is, by (9), equal to $\sum \lambda_s$, hence, by theorem 1, is not negative; and we find

$$0 \leq \sum_s (\lambda_s - \gamma_s u) < kn.$$

[source p. 675]

It follows from this, in view of (10), that a term of this sum, $\lambda_s - \gamma_s u$, cannot be smaller than $-(n - \delta_s)k$, for otherwise the total sum would be negative; and if, for simplicity, we take the lower bound somewhat smaller,

$$-nk\delta_s < \lambda_s - \gamma_s u < k\delta_s. \quad (11)$$

It is thereby proved that for every given system of numbers u, γ_s , provided only that condition (9) is fulfilled, there is an integer η of the field Ω which, with its conjugate numbers, satisfies the limiting conditions (11).

Now let g_s be a system of real quantities of which we demand only that

$$\gamma_1 g_1 + \gamma_2 g_2 + \cdots + \gamma_\nu g_\nu = \alpha$$

be different from zero. By (9) this excludes the case where all the g_s are equal to one another; but if they are not, then for every given system of the g_s there will be infinitely many systems γ_s that satisfy the requirement (9).

If now $k \sum \delta_s g_s = g$ is put, then from (11), by multiplication with g_s and summation, one obtains

$$-ng + \alpha u < \sum g_s \lambda_s < g + \alpha u. \quad (12)$$

According to this limiting condition, keeping g_s and α fixed, we now determine a sequence of integers

$$\eta, \eta', \eta'', \eta''', \dots, \quad (13)$$

by making, for u , a sequence of assumptions

$$u, u', u'', u''', \dots,$$

which are more precisely determined as follows.

First choose u so that

$$-ng + \alpha u = a, \quad g + \alpha u = a'$$

are positive; then put

$$\delta = \frac{a' - a}{\alpha} = \frac{(n+1)g}{\alpha},$$

and put

$$\begin{aligned} u' &= u + \delta, & u'' &= u' + \delta, & u''' &= u'' + \delta, & \dots, \\ a + \alpha\delta &= a', & a' + \alpha\delta &= a'', & a'' + \alpha\delta &= a''', & \dots, \end{aligned}$$

[source p. 676]

and from (12) we thus obtain, if $\lambda'_s, \lambda''_s, \dots$ are the conjugate logarithms of $\eta', \eta'', \dots$,

$$\begin{aligned} a &< \sum g_s \lambda_s < a', \\ a' &< \sum g_s \lambda'_s < a'', \\ a'' &< \sum g_s \lambda''_s < a''', \\ &\dots\dots\dots, \end{aligned} \quad (14)$$

and the sequence of these inequalities can be continued without bound.

Moreover all these numbers $\eta, \eta', \eta'', \dots$ have, by (6) and (7), the property that their absolute norms $N, N', N'', \dots$ remain below a fixed finite bound e^{nk} .

The number of possible values of $N, N', N'', \dots$ is therefore finite, certainly not greater than the greatest integer contained in e^{nk} . Likewise the number of all possible residues of the numbers $x_1, x_2, \dots, x_n$ [in (4)] modulo all the moduli $N, N', N'', \dots$ is finite; and from this it follows that, if we continue the sequence of numbers (13) far enough, two distinct numbers $\eta^{(h)}, \eta^{(k)}$ must occur in the sequence for which $N^{(h)} = N^{(k)}$ and the residues of $x_1, x_2, \dots, x_n$ modulo $N^{(h)}$ are exactly the same.

But from (14), if $h > k$ is assumed, it follows that

$$\sum g_s (\lambda_s^{(h)} - \lambda_s^{(k)}) > 0. \quad (15)$$

If N is the absolute norm of these two numbers $\eta^{(h)}, \eta^{(k)}$, then, because of the agreement of the residues of the x , the difference $\eta^{(h)} - \eta^{(k)}$ is divisible by N . On the other hand, by § 138 (13), the number N is divisible by $\eta^{(h)}$, and consequently $\eta^{(h)} - \eta^{(k)}$ is also divisible by $\eta^{(h)}$. It follows that $\eta^{(k)}$ is also divisible by $\eta^{(h)}$, and likewise $\eta^{(h)}$ by $\eta^{(k)}$. The two numbers are therefore associated, and if we put

$$\frac{\eta^{(h)}}{\eta^{(k)}} = \varepsilon,$$

then ε is a unit of the field Ω .

Moreover, if, denoting by $|\varepsilon_s|$ the absolute value of ε_s , we put

$$\delta_s \log |\varepsilon_s| = l_s, \quad (16)$$

then from (15) we obtain

$$g_1 l_1 + g_2 l_2 + \cdots + g_\nu l_\nu > 0. \quad (17)$$

[source p. 677]

This proves the theorem:

4. If $g_1, g_2, \dots, g_\nu$ is any system of real numbers which are not all equal to one another, then in every algebraic field Ω one can determine a unit ε such that the sum

$$g_1 l_1 + g_2 l_2 + \cdots + g_\nu l_\nu$$

is different from zero, where l_s is the system of conjugate logarithms of ε .

This important theorem, which is the foundation for what follows, is due, though in a somewhat changed form, to Dirichlet. This formulation was given by Minkowski.

§ 186. Systems of independent units and exponent systems of the units.

If in the theorem just proved we put $g_1 = 1, g_2 = 0, \dots, g_\nu = 0$, which is allowed if from now on we assume $\nu > 1$, it follows that in Ω there is a unit ε for which one of the conjugate logarithms, l_1 , is different from zero. Since every power of ε has the same property, there are also infinitely many such units.

This theorem now permits a very important generalization:

5. If $s \leq \nu - 1$, then in the field Ω one can determine a system of units with the associated conjugate logarithms

$$\begin{array}{ll} \varepsilon_1 : & l_{1,1}, \quad l_{1,2}, \quad \dots, l_{1,\nu}, \\ \varepsilon_2 : & l_{2,1}, \quad l_{2,2}, \quad \dots, l_{2,\nu}, \\ & \dots\dots\dots \\ \varepsilon_s : & l_{s,1}, \quad l_{s,2}, \quad \dots, l_{s,\nu}, \end{array} \quad (1)$$

in such a way that any one of the s -rowed determinants of the matrix of the $l_{h,k}$, for instance

$$L_s = \sum \pm l_{1,1} l_{2,2} \cdots l_{s,s},$$

is different from zero.

For $s = 1$ the stated theorem is correct by the remark made at the beginning. We therefore assume that it

[source p. 678]

has been proved for $s - 1$ and derive it generally by complete induction. For this purpose we arrange the determinant L_s according to the elements of the last row, and write it thus:

$$L_s = g_1 l_{s,1} + g_2 l_{s,2} + \cdots + g_s l_{s,s}, \quad (2)$$

where $g_s = L_{s-1}$ and hence, by the assumption made, can be taken to be different from zero. If we substitute these values of g in the formula of theorem 4, § 185, taking $g_{s+1} = 0, \dots, g_\nu = 0$, while g_s is not equal to zero, then, as long as $s < \nu$, certainly not all $g_1, g_2, \dots, g_\nu$ are equal to one another; and it follows that ε_s can be so chosen that L_s is different from zero, as was to be proved.

This mode of inference cannot be extended to $s = \nu$, and the determinant L_ν must indeed always be zero, since for every unit $\sum_1^\nu l_s$ vanishes.

For the case $s = \nu - 1$ we shall now denote the determinant L_s by

$$L_{\nu-1} = L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}), \quad (3)$$

and call a system of units $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ for which this determinant is different from zero a system of independent units.

The absolute value L of the determinant $L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1})$ is called the regulator of the system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$.⁵

If in the determinant

$$\begin{vmatrix} l_{1,1} & l_{1,2} & \cdots & l_{1,\nu} \\ \cdots & \cdots & \cdots & \cdots \\ l_{\nu-1,1} & l_{\nu-1,2} & \cdots & l_{\nu-1,\nu} \\ x_1 & x_2 & \cdots & x_\nu \end{vmatrix}, \quad (4)$$

in which $x_1, x_2, \dots, x_\nu$ are arbitrary quantities, we add all columns to the last one, then, with regard to the relations $\sum_1^\nu l_{s,i} = 0$, we obtain its value equal to

$$\pm(x_1 + x_2 + \cdots + x_\nu)L.$$

If, therefore, we take all the x except for an arbitrary one

[source p. 679]

to be $= 0$, and the remaining one to be $= 1$, then from (3) there results one of the $(\nu - 1)$ -rowed determinants of the matrix

$$\begin{matrix} l_{1,1} & \cdots & l_{1,\nu} \\ \cdots & \cdots & \cdots \\ l_{\nu-1,1} & \cdots & l_{\nu-1,\nu} \end{matrix}$$

and all these determinants therefore have, apart from the sign, the same value. It is therefore immaterial which one of the ν conjugate values is omitted in forming the regulator L .

If $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ is an independent system of units, and if $\lambda_1, \lambda_2, \dots, \lambda_\nu$ are the conjugate logarithms of any unit ε in Ω , then the numbers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ can always and uniquely be determined so that

$$\begin{aligned} \xi_1 l_{1,1} + \xi_2 l_{2,1} + \cdots + \xi_{\nu-1} l_{\nu-1,1} &= \lambda_1, \\ \xi_1 l_{1,2} + \xi_2 l_{2,2} + \cdots + \xi_{\nu-1} l_{\nu-1,2} &= \lambda_2, \\ &\dots\dots\dots \\ \xi_1 l_{1,\nu} + \xi_2 l_{2,\nu} + \cdots + \xi_{\nu-1} l_{\nu-1,\nu} &= \lambda_\nu \end{aligned} \quad (5)$$

⁵Dirichlet-Dedekind, *Vorlesungen über Zahlentheorie*, 4th ed., § 183.

holds. For the first $\nu - 1$ of these equations form a system of linear equations with non-vanishing determinant, and the ν th equation follows from the others because the sum of the conjugate logarithms of every unit vanishes.

We call the system of numbers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ the exponent system of the unit ε with respect to the system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$, and it remains to prove:

6. That the exponents of every unit ε are rational numbers whose denominators do not exceed a certain finite bound, and which therefore can all be assumed to have a common denominator depending only on the system ε_i .

To see this, one must consider the following:

1) If we regard the quantities $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ as variables and let each of them, independently of the others, run from 0 to 1, then the left sides of (5) remain enclosed within finite bounds depending only on the units $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$. Within these bounds, therefore, the conjugate logarithms λ of the unit ε must also remain, as long as

[source p. 680]

the exponents ξ_i remain restricted to non-negative proper fractional values; and hence, from the definition of the conjugate logarithms, one obtains an upper bound for ε and its conjugates.

But by theorem 2, § 185, there are in Ω only finitely many integers, and therefore still more only finitely many units, which in absolute value, with all their conjugates, lie below a finite bound.

If we now call a unit whose exponents lie between the bounds 0 and 1, including the lower and excluding the upper bound, a unit reduced with respect to the system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$, then we have the theorem:

There are only finitely many units in Ω which are reduced with respect to an independent system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$.

2) The units reproduce themselves by multiplication and division. If

$$\begin{array}{c} \xi'_1, \xi'_2, \dots, \xi'_{\nu-1}, \\ \xi''_1, \xi''_2, \dots, \xi''_{\nu-1} \end{array}$$

are the exponents of two units $\varepsilon', \varepsilon''$, then the sums and the differences

$$\xi'_1 \pm \xi''_1, \xi'_2 \pm \xi''_2, \dots, \xi'_{\nu-1} \pm \xi''_{\nu-1}$$

are the exponents of the units $\varepsilon'\varepsilon''$ and $\varepsilon' : \varepsilon''$.

This follows immediately from the theorem that the logarithm of a product or of a quotient is equal to the sum or the difference of the logarithms of the two components.

3) The units $\varepsilon_1, \varepsilon_2, \dots$ themselves have the exponents

$$1, 0, \dots, 0; \quad 0, 1, \dots, 0; \quad \dots,$$

and from 2) it follows that every system of integers, substituted for $\xi_1, \xi_2, \dots, \xi_{\nu-1}$, gives the exponent system of a unit which can be formed by multiplication of powers of the $\varepsilon_1, \varepsilon_2, \dots$

4) From 2) and 3) it follows that an exponent system $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ of a unit remains an exponent system if all its elements are multiplied by one and the same integer, and that it also retains this character when its elements ξ_i are increased or decreased by arbitrary integers. Thus, if we use the sign (x) in the sense of § 182, that it denotes the excess

[source p. 681]

of the number x over the greatest integer contained in x , the following is obtained:

If $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ is the exponent system of a unit, and m is an integer, then also

$$(m\xi_1), (m\xi_2), \dots, (m\xi_{\nu-1}) \quad (6)$$

is the exponent system of a unit.

Now the quantities $(m\xi_i)$, when they are not zero, are positive proper fractions; and by 1), if the sequence of integers is continued sufficiently far, it must therefore occur that for two distinct values m', m'' of m the number sequence (6) agrees. Since accordingly $m'\xi_i, m''\xi_i$ have the same excess over an integer, $(m' - m'')\xi_i$ is itself an integer; and so there is a rational integer m , at most equal to the number of exponent systems of reduced units, with the property that

$$m\xi_1, m\xi_2, \dots, m\xi_{\nu-1}$$

become rational integers.

This number m may change when the unit ε is changed. But since all possible values of this number lie below a finite bound, there is a finite number in which all these values of m are contained and which itself can be taken for m . Hence there is a certain positive integer m which can be taken as denominator of all rational numbers which can occur as exponents of units. This proves theorem 6.

§ 187. Fundamental systems of units.

We now choose, instead of the system of independent units ε_i , another system whose conjugate logarithms may be

$$\lambda_{i,1}, \lambda_{i,2}, \dots, \lambda_{i,\nu} \quad i = 1, 2, \dots, \nu - 1. \quad (1)$$

By § 186, (5),

$$\lambda_{i,k} = \xi_{1,i}l_{1,k} + \xi_{2,i}l_{2,k} + \dots + \xi_{\nu-1,i}l_{\nu-1,k}, \quad (2)$$

$$i = 1, 2, \dots, \nu - 1; \quad k = 1, 2, \dots, \nu,$$

where $\xi_{1,i}, \xi_{2,i}, \dots, \xi_{\nu-1,i}$ denote the exponents of one of the new units with respect to the system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$, which we may assume to be rational numbers with the common denominator m .

[source p. 682]

But by the multiplication theorem for determinants,

$$\begin{vmatrix} \lambda_{1,1} & \dots & \lambda_{\nu-1,1} \\ \dots & \dots & \dots \\ \lambda_{1,\nu-1} & \dots & \lambda_{\nu-1,\nu-1} \end{vmatrix} = \begin{vmatrix} \xi_{1,1} & \dots & \xi_{1,\nu-1} \\ \dots & \dots & \dots \\ \xi_{\nu-1,1} & \dots & \xi_{\nu-1,\nu-1} \end{vmatrix} \begin{vmatrix} l_{1,1} & \dots & l_{\nu-1,1} \\ \dots & \dots & \dots \\ l_{1,\nu-1} & \dots & l_{\nu-1,\nu-1} \end{vmatrix}, \quad (3)$$

or, if we put

$$\Lambda_{\nu-1} = \sum \pm \lambda_{1,1} \lambda_{2,2} \dots \lambda_{\nu-1,\nu-1}$$

and observe that the determinant

$$\sum \pm \xi_{1,1} \xi_{2,2} \dots \xi_{\nu-1,\nu-1}$$

is a rational number with denominator $m^{\nu-1}$,

$$\Lambda_{\nu-1} = \frac{a}{m^{\nu-1}} L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}), \quad (4)$$

where a is a rational integer.

It follows that the newly introduced units form an independent system if and only if the determinant of the $\xi_{i,k}$, that is, the number a , is different from zero.

Now among a finite or infinite number of non-zero rational fractions with the same denominator there is always one of least absolute value, and consequently, by (4), we can choose the new system of independent units so that its regulator $\pm\Lambda_{\nu-1}$ becomes as small as possible. Such a system we call a fundamental system of units; and the minimum value of the regulator itself, that is, the regulator of a fundamental system, we call the regulator of the field. We now assume that our system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ itself is a fundamental system.

Under this assumption it can be proved that all exponents of units must be integers.

Indeed, if we assume that there exists a unit ε whose exponents determined by the formulae § 186, (5) are not all integers, then by § 186, 4) there is also a unit ε_0 whose exponents are proper fractions and do not all vanish.

If by $\lambda_1, \lambda_2, \dots, \lambda_{\nu-1}$ in the formulae (5), § 186, we understand the system of conjugate logarithms of ε_0 , then

$$L(\varepsilon_0, \varepsilon_2, \dots, \varepsilon_{\nu-1}) = \xi_1 L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}),$$

[source p. 683]

and if we therefore assume, which is plainly no restriction of generality, that ξ_1 is a non-zero proper fraction, then this formula contradicts the assumption that

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$$

is a fundamental system, because the determinant $L_{\nu-1}$ would be made smaller if ε_1 were replaced by ε_0 .

Once we have a fundamental system, we can take any system of rational integers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ as an exponent system and form a unit ε with these exponents:

$$\varepsilon = \varepsilon_1^{\xi_1} \varepsilon_2^{\xi_2} \cdots \varepsilon_{\nu-1}^{\xi_{\nu-1}}.$$

It remains to answer the question how far a unit is determined by its exponent system.

Suppose that $\varepsilon', \varepsilon''$ are two units with the same exponent system. Then the exponent system of the unit $\varepsilon' : \varepsilon'' = \rho$ consists entirely of zeros, and consequently the conjugate logarithms of ρ are all equal to zero. Therefore ρ is an integer with the property that the absolute values of all numbers conjugate with ρ are equal to 1, and hence, as proved in § 175, 2, it is a root of unity. Thus the following theorem is proved:

1. There is in the field Ω a system of $\nu - 1$ fundamental units

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}, \tag{5}$$

having the property that in the form

$$\varepsilon = \rho \varepsilon_1^{\xi_1} \varepsilon_2^{\xi_2} \cdots \varepsilon_{\nu-1}^{\xi_{\nu-1}} \tag{6}$$

all units of the field, each only once, are contained, when $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ run through all rational integers and ρ runs through all roots of unity present in Ω .

It is now easy to derive all other fundamental systems from one fundamental system, by putting in (6), for the exponents, $\nu - 1$ different systems of integers $\xi_{i,k}$ whose determinant is ± 1 . For if one puts

$$\varepsilon'_i = \rho_i \varepsilon_1^{\xi_{1,i}} \varepsilon_2^{\xi_{2,i}} \cdots \varepsilon_{\nu-1}^{\xi_{\nu-1,i}},$$

then from (3) it follows that

$$L(\varepsilon'_1, \varepsilon'_2, \dots, \varepsilon'_{\nu-1}) = \pm L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}).$$

[source p. 684]

Thus both determinants have the minimum value, and both systems are fundamental systems.

The case $\nu = 1$, which we excluded above, occurs, apart from the case of the rational field, in which only the two units ± 1 exist, in the case of the imaginary quadratic field. In this case, in the twentieth section, we represented the integers of the field in the form

$$\frac{x + y\sqrt{-m}}{2},$$

where $-m$ denotes a fundamental discriminant, so that $m \equiv 0$ or $\equiv 3 \pmod{4}$, and x, y are both even or both odd. We obtain the units if we solve the equation

$$x^2 + my^2 = 4$$

in all possible ways in rational integers. But this equation, if $m > 4$, has only the two solutions $x = \pm 2, y = 0$, and thus in these cases, as in the field of rational numbers, there are only the two units ± 1 . If $m = 4$, one finds the four units $\pm 1, \pm i$, and finally if $m = 3$, the six units

$$\pm 1, \quad \pm \frac{-1 + i\sqrt{3}}{2}, \quad \pm \frac{-1 - i\sqrt{3}}{2}.$$

In the next simplest case, $n = 2, \nu = 2$, that is, in the real quadratic field, the theory of units coincides with the theory of Pell's equation treated in § 127 of the first volume.

§ 188. Reduced numbers.

[source p. 684]

If α is any non-zero integral or fractional number of the field Ω , then, as with the units, we consider the system of conjugate logarithms of α , by which we understand, as in § 185, the real numbers

$$\lambda_1 = \delta_1 \log |\alpha_1|, \quad \lambda_2 = \delta_2 \log |\alpha_2|, \dots, \quad \lambda_\nu = \delta_\nu \log |\alpha_\nu|, \quad (1)$$

which we also denote by $\lambda_s(\alpha)$. If we bring in the fundamental system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ of units, then

[source p. 685]

we can form the linear equations, analogously to § 186, (5):

$$\begin{aligned} \xi_1 l_{1,1} + \xi_2 l_{2,1} + \dots + \xi_{\nu-1} l_{\nu-1,1} + \delta_1 \xi_\nu &= \lambda_1, \\ \xi_1 l_{1,2} + \xi_2 l_{2,2} + \dots + \xi_{\nu-1} l_{\nu-1,2} + \delta_2 \xi_\nu &= \lambda_2, \\ &\dots\dots\dots \\ \xi_1 l_{1,\nu} + \xi_2 l_{2,\nu} + \dots + \xi_{\nu-1} l_{\nu-1,\nu} + \delta_\nu \xi_\nu &= \lambda_\nu, \end{aligned} \quad (2)$$

whose determinant [§ 186, (4)] has the non-zero value

$$\begin{vmatrix} l_{1,1} & l_{2,1} & \dots & l_{\nu-1,1} & \delta_1 \\ l_{1,2} & l_{2,2} & \dots & l_{\nu-1,2} & \delta_2 \\ \dots & \dots & \dots & \dots & \dots \\ l_{1,\nu} & l_{2,\nu} & \dots & l_{\nu-1,\nu} & \delta_\nu \end{vmatrix} = nL(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}). \quad (3)$$

Accordingly the unknowns $\xi_1, \xi_2, \dots, \xi_{\nu-1}, \xi_\nu$ are determined uniquely from (2).

By adding the equations (2) one first obtains very simply

$$n\xi_\nu = \log N_a(\alpha), \quad (4)$$

and therefore ξ_ν remains the same for all numbers α that are associated with one another.

The numbers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ are called the exponents of the number α [with respect to the fundamental system $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1})$].

From this definition it follows without further ado that the exponents of a product or of a quotient are equal to the sum or the difference of the corresponding exponents of the individual components. The exponents of a unit are rational integers, and every system of rational integers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ is also the exponent system of a unit.

The exponents of associated numbers therefore differ from one another by integers, and one can find, for each number α , an associated number α_0 whose exponents lie between 0 and 1. Such numbers are called **reduced numbers** (with respect to the fundamental system $\varepsilon_1, \dots, \varepsilon_{\nu-1}$). The reduced units are all, and only, the roots of unity present in Ω , whose number we shall denote by w . Since the two roots of unity ± 1 are contained in every field, w is at least $= 2$ and is always an even number.

If two associated numbers have the same exponent system, then they differ only by a factor which

[source p. 686]

is a root of unity. The exponent system of the reduced number α_0 derived from α is completely determined by α itself. This proves the theorem:

II. For every non-zero number α of the field Ω there exist w , and no more, reduced numbers $\rho\alpha_0$.

If another fundamental system is used in place of $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1})$, then the exponents $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ undergo an integral linear substitution of determinant ± 1 . A reduced number can then cease to be reduced for the new fundamental system. The concept of a reduced number is therefore dependent on the choice of the fundamental system. But the number of reduced numbers derived from a given number is always the same.

By this theorem we have attained the purpose of selecting from the whole infinite system of mutually associated numbers a definite finite number of representatives.

§ 189. Bounds for the number of integral numbers of the field Ω divisible by an ideal.

Our earlier considerations have shown that every integral number of a field Ω has only a finite number of ideals as divisors. Since every integral ideal is a divisor of its norm, there are consequently also only finitely many integral numbers in Ω whose absolute norm does not exceed a given bound.

From these numbers we now again pick out a part, and ask for the number T of all **integral numbers** of the field Ω whose absolute norm does not exceed a positive magnitude t , and which are divisible by a given ideal $\mathfrak{a}$.

If we take the rational field as a preliminary example, then there T is the number of natural numbers which are smaller than t and divisible by a given rational integer m , that is, in the notation used earlier (§ 182), $T = E\left(\frac{t}{m}\right)$.

[source p. 687]

The number T is, in general, not a function of t that can be determined more exactly; it grows to infinity together with t , but since it has only integral values it changes only step by step, and hence discontinuously. What matters here is the **limit to which the ratio $T : t$ tends when t grows without bound** (which in the example cited above is $1 : m$).

This limit can be found by considerations such as those which we have already used in § 155.

We first form a basis form of the given ideal $\mathfrak{a}$:

$$\alpha = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n, \quad (1)$$

that is, a linear function of the n variables x_i having the property that, by substituting all rational integers for the x_i , from α there arise all integral numbers of the field Ω that are divisible by $\mathfrak{a}$ (§ 146). By $\alpha_{i,1}, \alpha_{i,2}, \dots, \alpha_{i,n}$ we denote the values conjugate to α_i , and form the linear substitution for the x_i :

$$\begin{aligned} y_1 &= \alpha_{1,1}x_1 + \alpha_{2,1}x_2 + \cdots + \alpha_{n,1}x_n, \\ y_2 &= \alpha_{1,2}x_1 + \alpha_{2,2}x_2 + \cdots + \alpha_{n,2}x_n, \\ &\dots\dots\dots \\ y_n &= \alpha_{1,n}x_1 + \alpha_{2,n}x_2 + \cdots + \alpha_{n,n}x_n. \end{aligned} \quad (2)$$

If A is the determinant of this system of equations,

$$A = \sum \pm \alpha_{1,1} \alpha_{2,2} \cdots \alpha_{n,n},$$

then A^2 is the discriminant of the system of functions $(\alpha_1, \alpha_2, \dots, \alpha_n)$, and therefore, by § 147, is equal to the product of the square of the absolute norm of α and the field discriminant Δ . Since the absolute norm of α is at the same time the norm of the ideal $\mathfrak{a}$ (§ 151), we put

$$A^2 = N(\mathfrak{a})^2 \Delta. \quad (3)$$

Now we can, as in § 155, interpret $x_1, x_2, \dots, x_n$ as coordinates of a point (x) in a space of n dimensions, and let there correspond to each point (x) a point (x') whose coordinates are determined by

$$x'_1 = t^{\frac{1}{n}} x_1, \quad x'_2 = t^{\frac{1}{n}} x_2, \dots, \quad x'_n = t^{\frac{1}{n}} x_n, \quad (4)$$

where t denotes an arbitrary positive magnitude. To every region S of points (x) there corresponds a region S' of points (x') . All figures in S and the corresponding figures in S' are similar, and the linear dimensions in S are to those in S' as $1 : t^{\frac{1}{n}}$.

[source p. 688]

If we think of the region S as given, then the corresponding region S' will depend on t and will increase with t . The points with integral coordinates x_i we call, as before, lattice points. The number of lattice points lying in a finite region S' is always finite, but grows with t , and shall be denoted by Z_t . Then by the theorems of § 155, (14), the volume of the region S , that is, the n -fold integral extended over all points of S ,

$$\int \int \cdots \int dx_1 dx_2 \cdots dx_n,$$

is equal to the limit of the ratio $Z_t : t$ for t increasing without bound:

$$V = \lim_{t=\infty} \frac{Z_t}{t}. \quad (5)$$

This at the same time gives the more exact precision which we must impose on the region S if our consideration is to be applicable. The region S must be such that the n -fold integral V has a definite meaning. Every lattice point is now, by (1), the image of an integral number of the field Ω divisible by $\mathfrak{a}$, and thus we can also say that Z_t is the number of integral numbers in Ω divisible by $\mathfrak{a}$ whose images lie in the region S' .

In order to bound the region S' in a suitable way, we choose a system of fundamental units of the field Ω

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$$

with the system of conjugate logarithms

$$l_{i,1}, l_{i,2}, \dots, l_{i,\nu}$$

and with regulator

$$\pm L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}) = \pm \sum \pm l_{1,1} l_{2,2} \cdots l_{\nu-1,\nu-1}, \quad (6)$$

which may be denoted by L .

We now define a system of functions $\xi_1, \xi_2, \dots, \xi_\nu$ of the variables x_i by the following conditions.

Let, when the y_i have the meaning (2),

$$z_i = \delta_i \log |y_i|, \quad (7)$$

so that, when a system of rational integers is substituted for the x_i , z_i passes, by § 185, into the system of conjugate logarithms of the number α .

Then, if y_k belongs to a real field,

$$z_k = \frac{1}{2} \log y_k^2, \quad (8)$$

[source p. 689]

and if y_k, y'_k belong to an imaginary pair,

$$z_k = \log y_k y'_k, \quad (9)$$

whereby the definition is completely unambiguous.

We now determine the variables $\xi_1, \xi_2, \dots, \xi_{\nu-1}, \xi_\nu$ from the linear equations

$$\begin{aligned} \xi_1 l_{1,1} + \cdots + \xi_{\nu-1} l_{\nu-1,1} + \delta_1 \xi_\nu &= z_1, \\ \xi_1 l_{1,2} + \cdots + \xi_{\nu-1} l_{\nu-1,2} + \delta_2 \xi_\nu &= z_2, \\ &\dots\dots\dots \\ \xi_1 l_{1,\nu} + \cdots + \xi_{\nu-1} l_{\nu-1,\nu} + \delta_\nu \xi_\nu &= z_\nu, \end{aligned} \quad (10)$$

whose determinant does not vanish.

If integral values are given to the variables x_i , then by § 188, (2), (3) the $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ pass into the exponent system of the number α , and by addition of the equations (10) [§ 185, (3)] one obtains

$$\xi_\nu = \frac{1}{n} \log N_a(\alpha). \quad (11)$$

We now pass from the point (x) to the point (x') by putting $x'_i = t^{\frac{1}{n}} x_i$. Thereby y_i passes into

$$y'_i = t^{\frac{1}{n}} y_i,$$

and z_i into

$$z'_i = z_i + \frac{1}{n} \delta_i \log t.$$

If, therefore, we put z'_i in place of z_i , and denote by $\xi'_1, \xi'_2, \dots, \xi'_\nu$ the values of $\xi_1, \xi_2, \dots, \xi_\nu$ then obtained from (10), then

$$\xi'_1 = \xi_1, \quad \xi'_2 = \xi_2, \dots, \quad \xi'_{\nu-1} = \xi_{\nu-1}, \quad \xi'_\nu = \xi_\nu + \frac{1}{n} \log t. \quad (12)$$

We shall now bound the region S' by requiring that

$$0 \leq \xi'_1 < 1, \dots, 0 \leq \xi'_{\nu-1} < 1, \quad \xi'_\nu < \frac{1}{n} \log t \quad (13)$$

shall hold. In this way we achieve that the lattice points (x') lying in the region S' represent only reduced integral numbers α , and all and only those whose absolute norm is smaller than t .

Among the reduced numbers, however, there are always w associated with one another, if w denotes the number of roots of unity contained in Ω ; and if

[source p. 690]

we therefore combine these w associated numbers into one complex, then the number of these complexes is equal to the number T of all non-associated integral numbers in Ω divisible by $\mathfrak{a}$ whose absolute norm is smaller than t . Accordingly the number of lattice points lying in S' is

$$Z_t = wT. \quad (14)$$

For the bounding of the region S one obtains from (12) and (13):

$$0 \leq \xi_1 < 1, \dots, 0 \leq \xi_{\nu-1} < 1, \quad \xi_\nu < 0, \quad (15)$$

and from (5) we obtain

$$\lim_{t \rightarrow \infty} \frac{T}{t} = \frac{1}{w} \int \int \dots \int dx_1 dx_2 \dots dx_n = \frac{1}{w} V, \quad (16)$$

where the n -fold integral is to be extended over the region S determined by (15).

§ 190. Determination of the volume.

The volume V can be determined by the transformation formula for multiple integrals already adduced in § 155, according to which, if $x_1, x_2, \dots, x_n$ are functions of the variables $u_1, u_2, \dots, u_n$,

$$\begin{aligned} V &= \int \int \dots \int dx_1 dx_2 \dots dx_n \\ &= \int \int \dots \int \left(\sum \pm \frac{\partial x_1}{\partial u_1} \frac{\partial x_2}{\partial u_2} \dots \frac{\partial x_n}{\partial u_n} \right) du_1 du_2 \dots du_n, \end{aligned} \quad (1)$$

where the functional determinant in the integral taken with respect to the variables u is to be taken with its absolute value.

This functional determinant is also equal to the reciprocal value of the determinant

$$\sum \pm \frac{\partial u_1}{\partial x_1} \frac{\partial u_2}{\partial x_2} \dots \frac{\partial u_n}{\partial x_n}.$$

We first introduce the variables u by a linear substitution, and observe that among the fields conjugate with Ω , according to our assumption, there occur $n - \nu$ conjugate imaginary pairs

[source p. 691]

and $2\nu - n$ real ones. Accordingly, of the n functions y [§ 189, (2)]

$$y_k = \alpha_{1,k}x_1 + \alpha_{2,k}x_2 + \cdots + \alpha_{n,k}x_n \quad (2)$$

$2\nu - n$ are real and $2n - 2\nu$ are conjugate in pairs. By the n variables u we shall understand real linear combinations of the x , namely the $2\nu - n$ real y and the $2n - 2\nu$ components of the conjugate imaginary pairs of the y , so that, if y, y' is an imaginary pair,

$$2u_1 = y + y', \quad 2iu_2 = y - y',$$

or

$$y = u_1 + iu_2, \quad y' = u_1 - iu_2 \quad (3)$$

is put. If in the functional determinant of the y one applies twice, to each pair of conjugate imaginary rows, the elementary theorem on determinants concerning the addition of rows (Vol. I, § 22), then there results

$$\sum \pm \frac{\partial u_1}{\partial x_1} \frac{\partial u_2}{\partial x_2} \cdots \frac{\partial u_n}{\partial x_n} = \frac{1}{(-2i)^{n-\nu}} \sum \pm \frac{\partial y_1}{\partial x_1} \frac{\partial y_2}{\partial x_2} \cdots \frac{\partial y_n}{\partial x_n} = \frac{A}{(-2i)^{n-\nu}},$$

where A has the meaning fixed in § 189, (3).

The field discriminant is positive or negative according as the number of conjugate imaginary pairs, that is, $n - \nu$, is even or odd; consequently, if $\pm\Delta$ denotes the absolute value of the field discriminant, one obtains from § 189, (4), (16):

$$\lim \frac{T}{t} = \frac{2^{n-\nu}}{wN(\mathfrak{a})\sqrt{\pm\Delta}} \int \int \cdots \int du_1 du_2 \cdots du_n. \quad (4)$$

It therefore remains only to treat the integral

$$U = \int \int \cdots \int du_1 du_2 \cdots du_n.$$

From the equations § 189, (10), for every system of values of the variables $\xi_1, \xi_2, \dots, \xi_\nu$ satisfying the conditions (15), one obtains a definite finite system of values for $z_1, z_2, \dots, z_\nu$, and from these, by § 189, (7), for every y a non-zero absolute value. In the case of an imaginary pair (3), put

$$u_1 = e^{\frac{1}{2}z} \cos \vartheta, \quad u_2 = e^{\frac{1}{2}z} \sin \vartheta, \quad (5)$$

$$y_1 y'_1 = u_1^2 + u_2^2 = e^z. \quad (6)$$

[source p. 692]

Then, when ϑ is taken within the bounds

$$0 \leq \vartheta < 2\pi,$$

there results, for fixed z , a corresponding pair of values u_1, u_2 for each value of ϑ . Such an angle ϑ belongs to every imaginary pair.

Thus to every point of the region S there belongs one and only one system of values of the variables $\xi_1, \xi_2, \dots, \xi_\nu, \vartheta \dots$ within the bounds

$$0 \leq \xi_1 < 1, \dots, 0 \leq \xi_{\nu-1} < 1, \quad \xi_\nu < 0, \quad 0 \leq \vartheta < 2\pi, \dots \quad (7)$$

Conversely, if any system of values of the variables ξ, ϑ is given within the bounds (7), then from it one obtains uniquely the corresponding components u_1, u_2 of an imaginary y , but of the real y only the absolute values; hence to each real y either of the two signs can still be given. The number of these possible determinations is $2^{2\nu-n}$. If one of these determinations is selected, the variables x are uniquely determined and are finite for every finite system of values of the ξ, ϑ . For ξ_ν decreasing toward $-\infty$, the values of the y will in part converge to zero.

Accordingly the region S decomposes into $2^{2\nu-n}$ subregions $S_1, S_2, \dots$, each of which is characterized by a definite combination of signs of the real $y_1, y_2, \dots$; and accordingly the integral U decomposes into just as many partial integrals $U_1, U_2, \dots$,

$$U = U_1 + U_2 + \dots$$

In carrying out the integration in one of these components, which all turn out to be of the same value, we begin with an imaginary pair, if one is present.

If we regard u_1, u_2 as rectangular coordinates in a plane, then by (5) we can regard $e^{\frac{1}{2}z}, \vartheta$ as polar coordinates, and obtain

$$\int \int du_1 du_2 = \frac{1}{2} \int \int e^z dz d\vartheta,$$

where the integration with respect to ϑ , which extends from 0 to 2π , can be carried out. Thus one finds

$$\int \int du_1 du_2 = \pi \int e^z dz.$$

[source p. 693]

One now changes the order of integration and puts first any second imaginary pair that may still be present, which is again treated in the same way.

The different components $U_1, U_2, \dots$ differ by the signs of the u corresponding to the real y . We take the sign so that $\pm u$ becomes positive, and put

$$z = \log(\pm u), \quad \pm du = e^z dz. \quad (8)$$

Thus for the different $U_1, U_2, \dots$ we obtain the same expression by a ν -fold integral,

$$U_1 = \pi^{n-\nu} \int \int \dots \int e^{z_1+z_2+\dots+z_\nu} dz_1 dz_2 \dots dz_\nu, \quad (9)$$

if we take $dz_1, dz_2, \dots, dz_\nu$ to be positive.

But by (6) and (8) the variables z_k are here no others than those in § 189, (8), (9), (10), and hence

$$z_1 + z_2 + \dots + z_\nu = n\xi_\nu.$$

If we now apply the transformation formula (1) to the ν -fold integral (9), in order to introduce the variables $\xi_1, \xi_2, \dots, \xi_\nu$, then we have to form the determinant

$$\sum \pm \frac{\partial z_1}{\partial \xi_1} \frac{\partial z_2}{\partial \xi_2} \dots \frac{\partial z_\nu}{\partial \xi_\nu},$$

which by § 188, (3), agrees in absolute value with nL , that is, with the n -fold value of the field regulator L ; and therefore

$$U_1 = \pi^{n-\nu} nL \int_0^1 d\xi_1 \cdots \int_0^1 d\xi_{\nu-1} \int_{-\infty}^0 e^{n\xi_\nu} d\xi_\nu = \pi^{n-\nu} L$$

and

$$U = 2^{2\nu-n} \pi^{n-\nu} L.$$

Finally, therefore, (4) gives the theorem:

- 1. If T denotes the number of non-associated integral numbers divisible by $\mathfrak{a}$ whose absolute norm is smaller than t , then**

$$\lim_{t=\infty} \frac{T}{t} = \frac{2^\nu \pi^{n-\nu} L}{wN(\mathfrak{a})\sqrt{\pm\Delta}} = \frac{g}{N(\mathfrak{a})}, \quad (10)$$

where g is a positive number completely determined by the nature of the field Ω , namely:

$$g = \frac{2^\nu \pi^{n-\nu} L}{w\sqrt{\pm\Delta}}. \quad (11)$$

§ 191. Theorems from the theory of series.

[source p. 694]

For the further applications of the preceding results, some theorems from the theory of infinite series are required; these shall first be derived here.

- 1.** Let $a_1, a_2, a_3, \dots, a_n, \dots$ be an unbounded system of real quantities (positive, negative, or also vanishing), of which we suppose that the sum

$$\sigma_n = a_1 + a_2 + \cdots + a_n \quad (1)$$

remains, for every arbitrary n , below a finite bound C in absolute value. Then the series

$$S = \frac{a_1}{1^s} + \frac{a_2}{2^s} + \frac{a_3}{3^s} + \cdots \quad (2)$$

is, for every positive s , not only convergent, but also a continuous function of s .

To prove this theorem, we decompose S in this way:

$$S = S_m + R_m,$$

where

$$S_m = \frac{a_1}{1^s} + \frac{a_2}{2^s} + \cdots + \frac{a_{m-1}}{(m-1)^s},$$

$$R_m = \frac{a_m}{m^s} + \frac{a_{m+1}}{(m+1)^s} + \frac{a_{m+2}}{(m+2)^s} + \cdots.$$

Now by (1) one has

$$a_m = \sigma_m - \sigma_{m-1}, \quad a_{m+1} = \sigma_{m+1} - \sigma_m, \quad a_{m+2} = \sigma_{m+2} - \sigma_{m+1}, \dots$$

and therefore

$$R_m + \frac{\sigma_{m-1}}{m^s} = \sigma_m \left(\frac{1}{m^s} - \frac{1}{(m+1)^s} \right) + \sigma_{m+1} \left(\frac{1}{(m+1)^s} - \frac{1}{(m+2)^s} \right) + \dots$$

Since, by the assumption, $\sigma_{m-1}, \sigma_m, \sigma_{m+1}, \dots$ are enclosed between finite bounds $\pm C$, it follows that $R_m + \frac{\sigma_{m-1}}{m^s}$ is enclosed between the two bounds

$$\pm C \left(\frac{1}{m^s} - \frac{1}{(m+1)^s} + \frac{1}{(m+1)^s} - \frac{1}{(m+2)^s} + \dots \right) = \pm \frac{C}{m^s}$$

[source p. 695]

and hence, in absolute value,

$$R_m < \frac{2C}{m^s},$$

or, if $s > c$,

$$R_m < \frac{2C}{m^c}.$$

This upper bound for R_m , which is independent of s , can however, if c is positive, be depressed below any arbitrarily small value by taking m sufficiently large.

From this follows not only the convergence, but also the continuity of S . For if one takes m sufficiently large, then not only R_m , but also the fluctuation of R_m for variable s , becomes infinitely small, and S_m is, for a fixed m , a continuous function of s . Thus S too is continuous.

It is also to be observed that the assumption on σ_n by no means presupposes the convergence of the infinite series $\sum a_k$. It is not even necessary that the a_k be real; for if they are imaginary, one need only apply the theorem to the real and to the imaginary component. Finally it is also not necessary to assume $a_1, a_2, a_3, \dots$ to be constants. Everything remains valid when they are continuous functions of s .

2. Let $\mu_1, \mu_2, \dots, \mu_n, \dots$ denote an infinite set of positive numbers of such a kind that two finite numbers α, β can be given so that for every however large n

$$\alpha < \frac{n}{\mu_n} < \beta, \quad (3)$$

then the infinite series

$$S = \frac{1}{\mu_1^s} + \frac{1}{\mu_2^s} + \frac{1}{\mu_3^s} + \dots$$

is convergent so long as the exponent s is greater than 1, and the product $(s-1)S$, as s approaches the value 1 sufficiently closely, likewise remains between the bounds α and β , or at least approaches one of these values without bound.

[source p. 696]

The proof is very simple to conduct by reduction to an integral. For plainly

$$\frac{1}{(n+1)^s} < \int_n^{n+1} \frac{dx}{x^s} < \frac{1}{n^s},$$

because the first or the second value, $(n+1)^{-s}$ or n^{-s} , is obtained when, under the integral sign, the too small or the too large value $(n+1)^{-s}$ or n^{-s} is put for x^{-s} .

Thus from (3), for every positive s , one obtains

$$\alpha^s \int_n^{n+1} \frac{dx}{x^s} < \frac{1}{\mu_n^s} < \beta^s \int_{n-1}^n \frac{dx}{x^s}.$$

If we put

$$R_m = \frac{1}{\mu_{m+1}^s} + \frac{1}{\mu_{m+2}^s} + \frac{1}{\mu_{m+3}^s} + \cdots,$$

then this gives

$$\alpha^s \int_{m+1}^{\infty} \frac{dx}{x^s} < R_m < \beta^s \int_m^{\infty} \frac{dx}{x^s}$$

or

$$\frac{\alpha^s}{(s-1)(m+1)^{s-1}} < R_m < \frac{\beta^s}{(s-1)m^{s-1}}. \quad (4)$$

Thus R_m , however many terms may be summed, remains finite if $s > 1$, and for series with positive terms this is a sufficient condition for convergence.

If one further puts (4) in the form

$$\frac{\alpha^s}{(m+1)^{s-1}} < (s-1)R_m < \frac{\beta^s}{m^{s-1}},$$

one sees that the two bounds can be brought arbitrarily close to α, β if one takes $s-1$ small enough, and hence it follows that $(s-1)R_m$, for sufficiently small $s-1$, cannot lie outside the bounds α, β unless, as $s-1$ decreases, it approaches one of these values without bound. Since now R_m differs from S only by a component which is finite for every positive s , the theorem is thereby proved.

The theorem remains valid also when the inequality (3) does not hold for all n , but only holds as soon as n has exceeded an arbitrarily assigned bound m . On

[source p. 697]

the basis of this remark, if $n : \mu_n$ approaches a finite limiting value γ , one may take α and β arbitrarily close to this limiting value, and in this way obtains the following somewhat more precise form of the theorem:

3. If the positive numbers $\mu_1, \mu_2, \mu_3, \dots$ are so constituted that

$$\lim_{n \rightarrow \infty} \frac{n}{\mu_n} = \gamma$$

is a finite limiting value, then the infinite series

$$S = \frac{1}{\mu_1^s} + \frac{1}{\mu_2^s} + \frac{1}{\mu_3^s} + \cdots$$

is convergent for every s greater than 1, and

$$\lim_{s \rightarrow 1} (s-1)S = \gamma.$$

Let now any system M of infinitely many positive numbers μ_n be given, and let us arrange them as

$$\mu_1 \leq \mu_2 \leq \mu_3 \leq \cdots \quad (M), \quad (5)$$

so that in the series (5) a smaller term never follows a larger one; and let T denote the number of terms in M which are not greater than an arbitrarily given positive quantity t . Then the following theorem holds:

4. If one of the two limiting values

$$\lim_{n=\infty} \frac{n}{\mu_n}, \quad \lim_{t=\infty} \frac{T}{t} \quad (6)$$

is finite, then the other has the same finite value.

First let

$$\lim_{n=\infty} \frac{n}{\mu_n} = \gamma \quad (7)$$

be a finite limiting value. Then the μ_n must necessarily grow to infinity with infinitely increasing n , and for every positive t there is a value m such that

$$\mu_m \leq t < \mu_{m+1}; \quad (8)$$

m grows to infinity at the same time as t , and $T = m$ [by (5)]. Therefore by (8)

$$\frac{m}{\mu_m} \geq \frac{T}{t} > \frac{m}{\mu_{m+1}}. \quad (9)$$

[source p. 698]

If now $\gamma = 0$, then from this, letting t and m grow to infinity, it follows that $T : t$ also has the limiting value 0. But if γ is different from 0, then by (7)

$$\lim \frac{\mu_m}{\mu_{m+1}} = \lim \frac{m}{m+1} = 1,$$

and therefore the two limiting values in (9) approach the value γ , and hence it follows that

$$\lim_{t=\infty} \frac{T}{t} = \gamma. \quad (10)$$

Conversely, suppose secondly that the limit equation (10) is given. Then here too μ_n must grow to infinity with n , for otherwise, since the total number of all μ is infinite, T would already be infinite for a finite t , and γ could not be finite.

Now take a value μ_n which occurs in the series of mutually equal values

$$\mu_{m+1}, \mu_{m+2}, \dots, \mu_{m+l}$$

so that

$$m+1 \leq n \leq m+l.$$

Then T , when t passes through the value μ_n , will suddenly increase by l units, and the ratio $T : t$ increases by $l : \mu_n$. But since $T : t$ is to have a finite limiting value, one must have

$$\lim \frac{l}{\mu_n} = 0. \quad (11)$$

If now $t = \mu_n$, then $T = m + l$, and consequently

$$\frac{T}{t} = \frac{m+l}{\mu_n}, \quad \lim \frac{m+l}{\mu_n} = \gamma. \quad (12)$$

Further,

$$\frac{m}{\mu_n} < \frac{n}{\mu_n} \leq \frac{m+l}{\mu_n},$$

and consequently, by (11) and (12),

$$\lim \frac{n}{\mu_n} = \lim \frac{m}{\mu_n} = \gamma.$$

Thus equation (7) is derived from equation (10)⁶.

Using theorem 4, we can now also give theorem 3 the form:

[source p. 699]

5. If T is the number of quantities μ_n that are not greater than t , then

$$\lim_{s=1} \sum_{n=1}^{\infty} \frac{s-1}{\mu_n^s} = \lim_{t=\infty} \frac{T}{t}, \quad (13)$$

provided that the limiting value of $T : t$ is finite.

Assume that the quantities $\mu_1, \mu_2, \mu_3, \dots$ satisfy not only the conditions of theorem 3, but also the further conditions that, when

$$\mu_n = \frac{n + c_n}{\gamma} \quad (14)$$

is put, the c_n do not become infinite with infinitely increasing n (that is, for every however large n they are enclosed between finite bounds). Then theorem 3 can be replaced by the following sharper one:

6. If the numbers μ_n have the property that a finite γ can be determined so that

$$\gamma\mu_n - n = c_n \quad (15)$$

does not become infinite with infinitely increasing n , then the function of s defined for every $s > 1$ by

$$S = \frac{1}{\mu_1^s} + \frac{1}{\mu_2^s} + \dots \quad (16)$$

has the property that the difference

$$S - \frac{\gamma}{s-1} = C \quad (17)$$

approaches a finite limit as $s-1$ decreases without limit.

To prove this, put, by (14) and (16),

$$\sum_{n=1}^{\infty} \frac{\gamma^s}{n^s} - S = \gamma^s \sum_{n=1}^{\infty} \frac{1}{n^s} \left[1 - \left(1 + \frac{c_n}{n} \right)^{-s} \right]. \quad (18)$$

If now ε is a positive proper fraction, then, putting

$$a_n = \frac{1}{n^\varepsilon} \left[1 - \left(1 + \frac{c_n}{n} \right)^{-s} \right],$$

⁶This theorem is proved in essentially the same way in Dirichlet–Dedekind, Supplement II. Compare also Riemann’s mathematical works, second edition, No. XXX.

the product $a_n n^{1+\varepsilon}$ does not become infinite for infinitely increasing n , and consequently, by a known elementary theorem of the theory of series, $a_1 + a_2 + \dots$ is an unconditionally convergent series.

[source p. 700]

Now put

$$s = s_1 + \varepsilon.$$

Then from (18) one obtains

$$\sum \frac{\gamma^s}{n^s} - S = \gamma^s \sum \frac{a_n}{n^{s_1}},$$

and this, by 1., is a finite and continuous function of s_1 for positive s_1 . But for $s_1 = 1 - \varepsilon$ one has $s = 1$, and consequently

$$\sum \frac{\gamma^s}{n^s} - S = D_s \quad (19)$$

is finite for $s = 1$, namely equal to

$$D_1 = \sum \frac{\gamma^{c_n}}{n(n + c_n)}.$$

By application of simple theorems from the theory of the Γ -function, however, the sum $\sum \frac{1}{n^s}$ can be expressed by a definite integral. One has

$$\frac{1}{n^s} = \frac{1}{\Gamma(s)} \int_0^\infty e^{-nx} x^{s-1} dx,$$

and consequently

$$\sum \frac{1}{n^s} = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1} e^{-x} dx}{1 - e^{-x}}.$$

Further [by the theorem $(s-1)\Gamma(s-1) = \Gamma(s)$],

$$\frac{1}{s-1} = \frac{1}{\Gamma(s)} \int_0^\infty e^{-x} x^{s-2} dx,$$

and hence

$$\sum \frac{1}{n^s} - \frac{1}{s-1} = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} e^{-x} \left(\frac{1}{1 - e^{-x}} - \frac{1}{x} \right) dx.$$

Here the right-hand side is a continuous function for all positive values of s , and for $s = 1$ passes into Euler's constant

$$\int_0^\infty e^{-x} \left(\frac{1}{1 - e^{-x}} - \frac{1}{x} \right) dx = -\Gamma'(1) = 0.57721566 \dots$$

Now by (19)

$$D_s = \gamma^s \left(\sum \frac{1}{n^s} - \frac{1}{s-1} \right) - \left(S - \frac{\gamma^s}{s-1} \right),$$

[source p. 701]

and therefore

$$\lim_{s=1} \left(S - \frac{\gamma^s}{s-1} \right) = -\gamma\Gamma'(1) + D_1 = C_1, \quad (20)$$

which was to be proved, is finite⁷.

⁷The theorems on Γ -functions used here are found in the more detailed textbooks of integral calculus, for example Serret-Harnack, *Lehrbuch der Differential- und Integralrechnung*, vol. II, p. 170 f. (Leipzig 1885).

§ 192. Application to the determination of the class number.

We now apply these theorems to the theory of the field Ω . By the numbers μ_n of theorem 5 we understand the norms of all ideals of the field Ω , and therefore by T the number of ideals in Ω whose norm is not greater than a given positive quantity t ; and we obtain

$$\lim_{s=1} \sum \frac{s-1}{N(\mathfrak{a})^s} = \lim_{t=\infty} \frac{T}{t}, \quad (1)$$

where the sum on the left-hand side extends over all ideals $\mathfrak{a}$ of the field. Now by § 153 the ideals split into a finite number of classes

$$A_1, A_2, \dots, A_h,$$

and the great goal is the determination of this number h , the class number.

The ideals $\mathfrak{a}_1$ of one of these classes A_1 are characterized by the fact that their products with one and the same integral functional φ_1 , belonging to the class A_1^{-1} , are principal ideals, that is,

$$\varphi_1 \mathfrak{a}_1 = \alpha$$

is an integral number of the field Ω .

If the norm of $\mathfrak{a}_1$ is not greater than t , then

$$N_a(\alpha) \leq N_a(\varphi_1)t = t_1.$$

If T_1 is the number of ideals of the class A_1 whose norms are not greater than t , then T_1 is at the same time the number of non-associated integral numbers divisible by φ_1 whose norms are not

[source p. 702]

greater than t_1 ; and by the theorem § 190, 1,

$$\lim_{t_1=\infty} \frac{T_1}{t_1} = \frac{g}{N_a(\varphi_1)}, \quad (2)$$

where g has the meaning given at the cited place, and is therefore a non-zero number determined by the nature of the field Ω .

If now $T_2, t_2, \dots, T_h, t_h$ have the corresponding meaning for the classes $A_2, \dots, A_h$, as T_1, t_1 have for A_1 , then

$$T = T_1 + T_2 + \dots + T_h, \\ \frac{T}{t} = \frac{T_1}{t_1} N_a(\varphi_1) + \frac{T_2}{t_2} N_a(\varphi_2) + \dots + \frac{T_h}{t_h} N_a(\varphi_h),$$

and from (1) and (2) there results the fundamental formula

$$\lim_{s=1} \sum \frac{s-1}{N(\mathfrak{a})^s} = gh. \quad (3)$$

The number g may be regarded as known by § 185, (11), even though its actual calculation still rests on the assumption that a fundamental system of units is known, and even though the determination of such a system in higher fields has always been regarded as one of the greatest difficulties. Then the calculation of the class number still depends on determining the limiting value on the left-hand side of (3), which of course succeeds only in special cases,

but often permits important conclusions concerning the nature of the class number. We shall still develop a transformation of the sum

$$\sum \frac{1}{N(\mathfrak{a})^s} = \Phi(s) \quad (4)$$

into an infinite product, preparatory to the calculation.

Let $\mathfrak{p}$ be any prime ideal of f -th degree in the field Ω , so that

$$N(\mathfrak{p}) = p^f.$$

Then the infinite geometric series is

$$1 + \frac{1}{N(\mathfrak{p})^s} + \frac{1}{N(\mathfrak{p})^{2s}} + \cdots = \frac{1}{1 - p^{-sf}};$$

if these series are multiplied together for all distinct prime ideals $\mathfrak{p}$, then, by known theorems from

[source p. 703]

the theory of infinite series, one obtains a sum of terms of the form

$$\left(\frac{1}{N(\mathfrak{p})^x} \frac{1}{N(\mathfrak{p}')^{x'}} \frac{1}{N(\mathfrak{p}'')^{x''}} \cdots \right)^s = \left(\frac{1}{N(\mathfrak{p}^x \mathfrak{p}'^{x'} \mathfrak{p}''^{x''} \cdots)} \right)^s,$$

which are all contained in the form $N(\mathfrak{a})^{-s}$, and every such term arises once and only once. Thus

$$\Phi(s) = \prod_{\mathfrak{p}} \frac{1}{1 - N(\mathfrak{p})^{-s}}. \quad (5)$$

If therefore $\mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_e$ are the mutually distinct prime factors of p , and $f_1, f_2, \dots, f_e$ their degrees (§ 143), then

$$\Phi(s) = \prod_p \frac{1}{(1 - p^{-sf_1})(1 - p^{-sf_2}) \cdots (1 - p^{-sf_e})}, \quad (6)$$

where the infinite product $\prod$ is to be extended over all natural prime numbers p , and by (3), (4)

$$gh = \lim_{s=1} (s-1)\Phi(s). \quad (7)$$

From this one can draw a conclusion important for many applications:

If from the development

$$1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots = \frac{1}{1 - p^{-s}}$$

we form the product for all prime numbers p , then we get

$$\prod_p \frac{1}{1 - p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad (8)$$

where the sum on the right-hand side extends over all natural numbers n ; and thus this product has a finite value for all values of s greater than 1. The reciprocal value, namely the product

$$P = \prod_p (1 - p^{-s}), \quad (9)$$

whose factors are all smaller than 1, nevertheless has, so long as $s > 1$, a value different from zero. This property is preserved if in forming the product P we consider only a part of all prime numbers p , since the product is only increased by omitting arbitrary factors, without ever exceeding unity.

[source p. 704]

It follows that in the product (6), the partial products

$$\prod \frac{1}{1 - p^{-sf}},$$

which extend over all prime numbers p for which $f > 1$, remain finite even for $s = 1$. Since, on the other hand, g, h have definite non-zero values, the following important theorem follows from (7):

I. If $\mathfrak{p}$ runs through the totality of prime ideals of first degree of any algebraic field Ω , then the product

$$(s - 1) \prod \frac{1}{1 - N(\mathfrak{p})^{-s}} \quad (10)$$

has, for $s = 1$, a finite limiting value different from zero.

This property is preserved also if, in forming the product, an arbitrary finite number of prime ideals of first degree is omitted.

The norms $N(\mathfrak{p})$ in formula (10) are equal to natural prime numbers p . But a prime number p occurs in it as often as it contains prime factors of first degree in Ω , hence at most n times. If Ω is a normal field, then every prime number p which is decomposable into prime factors of first degree actually occurs in it exactly n times, and for the product (10) we may also write

$$(s - 1) \prod \frac{1}{(1 - p^{-s})^n},$$

where the product $\prod$ extends over all prime numbers p decomposable into prime factors of first degree (§ 160).

An immediate consequence of theorem I is:

In every algebraic field there are infinitely many prime ideals of first degree.

Twenty-fourth Section.

Class number of the cyclotomic fields.

§ 193. Class-number representation in the cyclotomic field Ω_m .

[source p. 705]

The general results concerning the class number h shall now be applied to the full cyclotomic field Ω_m , under the standing assumption that m is a power of a prime number q , and hence that

$$m = q^\alpha, \quad \varphi(m) = q^{\alpha-1}(q - 1) = \mu \quad (1)$$

has been put.

In the twenty-first section (§ 169, 171) we saw that in q only one prime ideal of first degree occurs, and that a prime number p different from q and belonging to the exponent f decomposes into e distinct prime ideals of f -th degree. Here f denotes the smallest positive exponent for which

$$p^f \equiv 1 \pmod{m} \quad (2)$$

and e is determined by the equation

$$\mu = ef. \quad (3)$$

The function $\Phi(s)$ determined at the end of the preceding paragraph by formula (6) therefore here obtains the expression

$$\Phi(s) = \frac{1}{1 - q^{-s}} \prod \frac{1}{(1 - p^{-sf})^e}, \quad (4)$$

where the product sign $\prod$ extends over all prime numbers p different from q . Here s is a variable which is always greater than 1, and is finally to approach the bound 1.

[source p. 706]

In order now to transform this expression for the function Φ further and prepare it for calculation, one must use the theorems on Abelian groups and group characters which we learned in paragraphs 14 to 16 of this volume.

The totality of the numbers n not divisible by q , taken modulo m , forms an Abelian group $\mathfrak{R}$ of degree μ , whose characters $\chi(n)$ were determined in § 16. There was a small difference according as q is odd or $q = 2$.

1) If q is odd, then there is a primitive root c of m , and for every number n there is an index γ such that

$$n \equiv c^\gamma \pmod{m}. \quad (5)$$

If then Θ is a μ -th root of unity, then

$$\chi(n) = \Theta^\gamma, \quad (6)$$

and all μ characters are obtained by putting for Θ the different μ -th roots of unity.

If n belongs to the exponent f , then e is the greatest common divisor of γ and μ , and $\chi(n)$ is an f -th root of unity.

2) For $q = 2$, $m > 4$ (the case $m = 4$ is not considered here), every odd number n has two indices α, β , determined modulo $2, \frac{1}{2}\mu$, for which

$$n \equiv (-1)^{\alpha} 5^{\beta} \pmod{m} \quad (7)$$

holds. As characters one obtains, if $\varepsilon = \pm 1$ and Θ is put equal to a $\frac{1}{2}\mu$ -th root of unity,

$$\chi(n) = \varepsilon^{\alpha} \Theta^{\beta}; \quad (8)$$

f is the smallest positive solution of the two congruences

$$\alpha f \equiv 0 \pmod{2}, \quad \beta f \equiv 0 \pmod{\frac{1}{2}\mu},$$

and here also $\chi(n)$ is an f -th root of unity.

For both cases, however, the following remark holds:

3) If n belongs to the exponent f , then, when χ is allowed to run through the whole group of the μ characters, every f -th root of unity is obtained e times from $\chi(n)$.

For since the χ form an Abelian group, one first obtains each f -th root of unity which occurs at all equally often, namely as often as the value 1, since if $\chi_1(n) = \chi_2(n)$,

[source p. 707]

then $\chi_1\chi_2^{-1}(n) = \chi_0(n) = 1$. On the other hand, however, every f -th root of unity is obtained; for if in case 1) Θ is taken to be a primitive μ -th root of unity, and in case 2) a primitive $\frac{1}{2}\mu$ -th root of unity, and in the latter case at the same time $\varepsilon = -1$, then $\chi(n)$ is a primitive f -th root of unity, and from its powers all f -th roots of unity can be derived.

If now x denotes a variable, then, when n belongs to the exponent f , the product extended over all characters χ is

$$\prod_{\chi} [1 - \chi(n)x] = (1 - x^f)^e, \quad (9)$$

and if one therefore puts $x = p^{-s}$, then from (4) one obtains

$$\Phi(s) = \frac{1}{1 - q^{-s}} \prod_{\chi} \prod_p \frac{1}{1 - \chi(p)p^{-s}}, \quad (10)$$

where the first product sign extends over all characters χ , and the second over all prime numbers p . Thus $\Phi(s)$ is a product of a finite number, μ , of factors, each of which is an infinite product.

We now expand each factor of one of these infinite products in increasing powers of p^{-s} . This gives

$$\frac{1}{1 - \chi(p)p^{-s}} = 1 + \chi(p)p^{-s} + \chi(p^2)p^{-2s} + \chi(p^3)p^{-3s} + \dots$$

The product of all factors so obtained, if χ is kept fixed while p runs through all primes different from q , is an aggregate of terms of the form

$$\chi(p^k)\chi(p'^{k'})\chi(p''^{k''}) \dots p^{-sk}p'^{-sk'}p''^{-sk''} \dots = \chi(n)n^{-s},$$

and every term of this form occurs in the product once and only once [compare § 192, (8)]. Thus the transformation of $\Phi(s)$ into a finite product of infinite series is obtained:

$$\Phi(s) = \frac{1}{1 - q^{-s}} \prod_{\chi} \sum_n \frac{\chi(n)}{n^s}, \quad (11)$$

where the sum extends over all numbers n not divisible by q , and the product over all μ characters χ .

This expression is suited to determine the limiting value of $(s - 1)\Phi(s)$ for $s = 1$. Let us first consider the sum corresponding to the principal character $\chi = 1$:

$$\sum_n \frac{1}{n^s}.$$

[source p. 708]

The numbers n are obtained by removing from the totality of all natural numbers k the multiples of q , that is, the totality of the numbers qk . Hence

$$\sum_n \frac{1}{n^s} = \sum_k \frac{1}{k^s} - \sum_k \frac{1}{q^s k^s} = (1 - q^{-s}) \sum_k \frac{1}{k^s}.$$

But if in theorem 3, § 191, we put for the μ_n the sequence of natural numbers k , then

$$\lim_{s=1} (s - 1) \sum_k \frac{1}{k^s} = 1,$$

and therefore we obtain

$$\lim_{s=1} \frac{s-1}{1-q^{-s}} \sum_n \frac{1}{n^s} = 1. \quad (12)$$

The other factors of the product $\Phi(s)$, however, are, when the infinite series are ordered according to increasing values of n , continuous functions of s by theorem 1, § 191.

For the sum ordered according to increasing values of n ,

$$\sum \chi(n), \quad (13)$$

is indeed not convergent, but nevertheless cannot go beyond a finite bound in absolute value. For as often as n runs through a complete residue system modulo m , a contribution is added to the sum (13) which, by § 11, 6, has the value 0. Hence, if χ is not the principal character,

$$\lim_{s=1} \sum_n \frac{\chi(n)}{n^s} = \sum_n \frac{\chi(n)}{n}, \quad (14)$$

and consequently, for the class number h in the cyclotomic field Ω_m , we now obtain the expression

$$gh = \prod_{\chi} \sum_n \frac{\chi(n)}{n}, \quad (15)$$

in which the product sign $\prod$ refers only to the characters χ different from the principal character.

§ 194. Determination of the sums X .

The infinite series

$$X = \sum_n \frac{\chi(n)}{n}, \quad (1)$$

[source p. 709]

on which, according to the preceding, the determination of the class number still depends, can be represented by finite expressions. Namely,

$$\frac{1}{n} = \int_0^1 x^{n-1} dx,$$

and therefore

$$X = \int_0^1 \sum_n \chi(n) x^{n-1} dx.$$

Now $\chi(n)$ remains unchanged when n is increased by a multiple of m . If, however, t denotes the least positive residue of n and one puts

$$n = t + ml,$$

then $\chi(n) = \chi(t)$, and it follows that

$$X = \int_0^1 \sum_t \chi(t) x^{t-1} \sum_{l=0}^{\infty} x^{ml} dx.$$

Put now

$$f(x) = \sum_t \chi(t) x^t. \quad (2)$$

Then $f(x)$ is an integral function of x of degree $m - 1$, and at the same time, by the relation $\sum \chi(t) = 0$,

$$f(0) = 0, \quad f(1) = 0, \quad (3)$$

so that $f(x)$ is divisible by $x(1 - x)$. For X , by the summation formula for the geometric series $\sum x^{ml}$, one obtains

$$X = \int_0^1 \frac{f(x) dx}{x(1 - x^m)}. \quad (4)$$

To find such an integral, integral calculus prescribes decomposing the rational fraction under the integral sign into partial fractions. This partial-fraction decomposition gives, however, by (3), if

$$r = e^{\frac{2\pi i}{m}}$$

is put:

$$\frac{f(x)}{x(1 - x^m)} = -\frac{1}{m} \sum_{s=1}^{m-1} \frac{f(r^s)}{x - r^s}$$

[Vol. I, § 29, (11)], and now we have further, by known

[source p. 710]

elementary theorems of integral calculus, if α denotes any angle between 0 and 2π ,

$$\int_0^1 \frac{dx}{x - e^{i\alpha}} = \frac{1}{2} \log \left(4 \sin^2 \frac{\alpha}{2} \right) + i \frac{\pi - \alpha}{2}.$$

It follows from this, by (4), that

$$X = -\frac{1}{m} \sum_{s=1}^{m-1} f(r^s) \left[\frac{1}{2} \log \left(4 \left(\sin \frac{s\pi}{m} \right)^2 \right) - \frac{i\pi(m - 2s)}{2m} \right]. \quad (5)$$

Here by (2)

$$f(r^s) = \sum_t \chi(t) r^{st}, \quad (6)$$

and hence [by (3)]

$$\sum_{s=1}^{m-1} f(r^s) = \sum_{s=0}^{m-1} f(r^s) = 0, \quad (7)$$

since

$$\sum_{s=0}^{m-1} r^{st} = 0.$$

Accordingly formula (5) simplifies somewhat further and gives

$$X = -\frac{1}{m} \sum_{s=1}^{m-1} f(r^s) \left[\frac{1}{2} \log \left(\sin \frac{s\pi}{m} \right)^2 + \frac{i\pi s}{m} \right], \quad (8)$$

for which one may also, replacing s by $m - s$ and again using (7), put

$$X = -\frac{1}{m} \sum_{s=1}^{m-1} f(r^{-s}) \left[\frac{1}{2} \log \left(\sin \frac{s\pi}{m} \right)^2 - \frac{i\pi s}{m} \right]. \quad (9)$$

From (6), however, follows

$$f(r^{-s}) = \sum_t \chi(t) r^{-st},$$

and the sum taken over t is not changed if t is replaced by $m - t$. Hence

$$f(r^{-s}) = \chi(-1)f(r^s).$$

The factor $\chi(-1)$ has the value $+1$ or -1 , since its square is $= +1$. We therefore now distinguish two kinds of characters $\chi_1(n), \chi_2(n)$, so that

$$\chi_1(-1) = 1, \quad \chi_2(-1) = -1 \quad (10)$$

[source p. 711]

and accordingly the functions f are also to be distinguished into f_1 and f_2 , so that

$$f_1(r^{-s}) = f_1(r^s), \quad f_2(r^{-s}) = -f_2(r^s). \quad (11)$$

Likewise we distinguish the expressions (8), (9) as X_1 and X_2 , and, adding the two expressions, we get by (11)

$$\begin{aligned} X_1 &= -\frac{1}{2m} \sum_{s=1}^{m-1} f_1(r^s) \log \left(\sin \frac{s\pi}{m} \right)^2, \\ X_2 &= -\frac{i\pi}{m^2} \sum_{s=1}^{m-1} s f_2(r^s). \end{aligned} \quad (12)$$